
C^* -Algebras and K-Theory

— *EYNTKA* Series —

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A C^* -algebra is, by the Gelfand–Naimark theorem, nothing other than a norm-closed $*$ -algebra of bounded operators on a Hilbert space; and by the commutative case of that theorem, a commutative C^* -algebra is nothing other than the algebra $C_0(X)$ of continuous functions vanishing at infinity on a locally compact Hausdorff space X . These two facts frame the whole subject. The second makes “commutative C^* -algebra” and “locally compact space” synonymous, so that a general, noncommutative C^* -algebra is to be read as a *noncommutative space*; the first guarantees the dictionary is concrete, every C^* -algebra being realized faithfully by operators. The theory of C^* -algebras is the geometry of noncommutative spaces, written in the language of operators.

The organizing question of this book is how the invariants of topology survive the passage to the noncommutative world, and the cleanest answer is K -theory. To a C^* -algebra A one attaches groups $K_0(A)$ and $K_1(A)$, built from the projections and unitaries of its matrix algebras; they are 2-periodic (a six-term exact sequence and Bott periodicity, exactly as for topological K -theory) and they package operator-theoretic data, most vividly the Fredholm index, as in the Toeplitz extension. The Serre–Swan theorem closes the circle: for X compact, projections over $C(X)$ are vector bundles over X , so $K_0(C(X)) = K^0(X)$, and the operator theory recovers the topological K -theory of [2] exactly.

This book develops C^* -algebras from their axioms through Gelfand–Naimark duality, states and the GNS construction, and representations, and then through the K -theory just described. It is the operator-algebraic member of the K -theory family, alongside [2] (topological) and [3] (algebraic). Beyond that role it is meant as a foundation: the index theory of elliptic operators, noncommutative geometry, and the harmonic analysis of group C^* -algebras all begin here.

We assume the functional analysis of [4] (Banach and Hilbert spaces, the spectral theorem for bounded operators) and the measure theory of [NateMeasureTheory]; no prior operator-algebra theory is presumed. A continuous function on a compact space and the Toeplitz algebra serve as the two examples to keep in mind throughout, the first commutative and geometric, the second noncommutative and analytic.

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Basics of C^ algebras*

C^* algebras where one of the heavily studied objects of K -theory, so a word must be said about them.

Definition 1.0.1: C^* -Algebra

A C^* -algebra A is an algebra over $\mathbb{C}$ with norm $a \mapsto \|a\|$ and involution $a \mapsto a^*$ such that A is complete with respect to the norm, $\|ab\| \leq \|a\|\|b\|$, and $\|a^*a\| = \|a\|^2$ for all $a, b \in A$

A C^* -algebra is called *unital* if it has a multiplicative identity. It is called *separable* if it contains a countable dense subset.

Given the axioms, it is easy to check that $\|a\| = \|a^*\|$.

Definition 1.0.2: $*$ -Homomorphism

Let A, B be C^* -algebras. Then a map $\varphi : A \rightarrow B$ is called a $*$ -homomorphism if it is a ring homomorphism that satisfies $\varphi(a^*) = \varphi(a)^*$. If A, B are unital and $\varphi(1_A) = 1_B$, then φ is called *unital* (or *unital preserving*)

A sub- C^* -algebra of A is a non-empty subset $B \subseteq A$ if it is closed under the 4 operations of A (addition, multiplication, adjoint, and scalar multiplication), as well as being a closed subset of A (so that it remains complete under the norm). If we simply have a sub $*$ -algebra of a C^* -algebra, then taking the norm closure gives us a sub C^* -algebra (since the 4 operations are continuous with respect to $\|\cdot\|$). Similar natural elaborations occur for generating C^* -algebra from a subset $S \subseteq A$. Quotients and products of C^* -algebras naturally follow the discussion from Banach Spaces (see Functional Analysis notes for details).

An important characterization of C^* -algebras is their “geometric” dual

Theorem 1.0.3: Gelfand-Naimark Duality

Let A be a C^* -algebra. Then there exists a Hilbert space H and an isometric $*$ -homomorphism $\varphi : A \rightarrow B(H)$. Thus every C^* -algebra is isomorphic to a sub C^* -algebra of $B(H)$. If A is separable, then H can be chosen to be a separable Hilbert Space

Proof:

see notes on functional Analysis, Gelfand Duality ([4]). The high-level idea is that we may view A as a vector-space equipped with a suitable inner product that comes from the positive linear functionals on A , which then we may complete. The rest is similar to Yoneda's Embedding theorem (recall the special case of a ring R being embedded in $B(R) = \text{End}_{\mathbf{Ab}}(R)$)

As mentioned, we shall consider C^* -algebras to not necessarily be unital. However, we may universally associate a unique unital C^* -algebra $\tilde{A}$ to a non-unital one A , with the characterization that $\tilde{A}/A \cong \mathbb{C}$.

Proposition 1.0.4: Unitization

Let A be a C^* -algebra. Then there exists a unital C^* -algebra $\tilde{A}$ such that $\tilde{A}/A \cong \mathbb{C}$. Furthermore, unitization is functorial.

Proof:

Let

$$\tilde{A} = A \oplus \mathbb{C}$$

with operation

$$(a, \alpha) \cdot (b, \beta) = (ab + \beta a + \alpha b, \alpha\beta) \quad (a, \alpha) = (a^*, \bar{\alpha})$$

and addition/scalar-multiplication is done coordinate wise. Then it is enough now to show that $(0, 1)$ is the unit in $\tilde{A}$. The rest of the properties now easily follow.

If A is already unital, then we get $\tilde{A} \cong A \oplus \mathbb{C}$ satisfies this property.

Every C^* -algebra naturally induces a new C^* -algebra $M_n(A)$ known as its *matrix algebra*. Each element of $M_n(A)$ is an $n \times n$ matrix whose elements are in A , addition, scalar multiplication, and involution is done term-wise, multiplication is matrix multiplication, and norm is the max of all the norms on the entries. With this norm, a function $f : X \rightarrow M_n(A)$ is continuous if and only if each component map $f_{ij} : X \rightarrow A$ is continuous. Naturally, if $\varphi : A \rightarrow B$ is a C^* -homomorphism, then $\varphi_n : M_n(A) \rightarrow M_n(B)$ given component wise operation is a C^* -homomorphism, and $M_n(-)$ is functorial.

Defining semi-group via Projections

First, we shall say two elements $a, b \in X$ are homotopic if there is a continuous path $[0, 1] \rightarrow X$ where $\gamma(0) = a$ and $\gamma(1) = b$ (this comes from the idea that X can be embedded in a natural Hilbert space $L^2(X)$)

Definition 1.0.5: Projections And Equivalences

Let A be a C^* -algebra. The set of all projection of A shall be denoted $\mathcal{P}(A)$ (set of all elements such that $p = p^2 = p^*$). We shall define two equivalence relations on $\mathcal{P}(A)$:

1. (*von Neumann equivalence*) $p \sim q$ if there exists $v \in A$ such that $p = v^*v$ and $q = vv^*$
2. (*unitary equivalence*) $p \sim_u q$ if there exists a unitary element $u \in \mathcal{U}(\tilde{A})$ where $q = upu^*$.

Definition 1.0.6: Set of projections

Let

$$\mathcal{P}(A) = \mathcal{P}(M_n(A)) \quad \mathcal{P}_\infty(A) = \bigcup_n^\infty \mathcal{P}_n(A)$$

Then define $\sim_0$ on $\mathcal{P}(A)$ as follows

$$p \sim_0 q \iff v \in M_{m,n}(A), p = v^*v \text{ and } q = vv^*$$

now define $\oplus$ on $\mathcal{P}_\infty(A)$ via:

$$p \oplus q = \text{diag}(p, q) = \begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix}$$

Then it is not hard to see this forms a semi-group structure

Proposition 1.0.7: Semigroup Of Projections

Let p, q, r, p', q' be projection in $\mathcal{P}_\infty(A)$ for some C^* -algebra A . Then:

1. $p \sim_0 p \oplus 0_n$ for every n
2. $p \sim_0 p'$ and $q \sim_0 q'$, then $p \oplus q \sim_0 p' \oplus q'$
3. $p \oplus q \sim_0 q \oplus p$
4. if p, q are projections in $\mathcal{P}_n(A)$ such that $pq = 0$, then $p+q$ is a projection and $p+q \sim_0 p \oplus q$
5. $(p \oplus q) \oplus r = p \oplus (q \oplus r)$

Proof:

all relatively simple, will leave for now

Definition 1.0.8: The Semigroup for C^* -algebras

Let $(\mathcal{P}(A), \sim_0, \oplus)$ be as in the above. Then

$$\mathcal{D}(A) = \mathcal{P}_\infty(A) / \sim_0$$

forms a semi-group where

$$[p]_{\mathcal{D}} + [q]_{\mathcal{D}} = [p \oplus q]_{\mathcal{D}}$$

Naturally, by the Grothendieck construction, this abelian group can be turned into a group. We shall use this in section 1.1 to define the K_0 group.

1.1 K_0 of a C^* -Algebra

The topological K -theory of [2] attaches to a compact Hausdorff space X the Grothendieck group of its complex vector bundles. The semigroup machinery of definitions 1.0.6 and 1.0.8 has already packaged the algebraic shadow of this construction: projections over a C^* -algebra play the role that vector bundles play over a space, and Murray–von Neumann equivalence (definition 1.0.5) plays the role of bundle isomorphism. This section completes the passage from the semigroup $\mathcal{D}(A)$ to a genuine abelian group, the K_0 -group of A , and develops its three structural pillars: functoriality together with homotopy invariance, stability under passing to matrix algebras and to the stabilization $A \otimes \mathcal{K}$, and the half-exact behaviour on short exact sequences of C^* -algebras. The payoff is a precise bridge between operator algebras and topology: for X compact Hausdorff the group $K_0(C(X))$ is exactly the topological $K^0(X)$ of [2]. This is the operator-algebraic incarnation of the Serre–Swan correspondence between vector bundles and finitely generated projective modules, and it is the reason the same letter K names two theories that at first sight live in different worlds.

The starting point is the following definition.

Definition 1.1.1: K_0 of a Unital C^* -Algebra

Let A be a unital C^* -algebra and let $\mathcal{D}(A) = \mathbb{CP}_\infty(A) / \sim_0$ be the abelian semigroup of definition 1.0.8, whose elements are the Murray–von Neumann equivalence classes $[p]_{\mathcal{D}}$ of projections over A under the block-diagonal sum $\oplus$. The K_0 -group of A is the Grothendieck group

$$K_0(A) := G(\mathcal{D}(A))$$

the universal abelian group receiving a semigroup homomorphism $\gamma: \mathcal{D}(A) \rightarrow K_0(A)$.

The definition packages $K_0(A)$, for unital A , as the Grothendieck group of the semigroup $\mathcal{D}(A) = \mathbb{CP}_\infty(A) / \sim_0$ of definition 1.0.8, whose elements are Murray–von Neumann equivalence classes of projections over A and whose addition is the block-diagonal sum $\oplus$. Recall (definition 1.0.5) that a projection is a self-adjoint idempotent $p = p^2 = p^*$, and that $p \sim_0 q$ when there is a rectangular partial isometry v over A with $p = v^*v$ and $q = vv^*$. The Grothendieck construction formally adjoins additive inverses to the classes $[p]_{\mathcal{D}}$, so that a general element of $K_0(A)$ is a formal difference of two classes of projections. The first task is to make this informal description into a usable normal form.

Proposition 1.1.2: Standard Picture of K_0

Let A be a unital C^* -algebra. Then every element of $K_0(A)$ has the form

$$[p] - [q]$$

for projections $p, q \in \mathbb{CP}_\infty(A)$, where $[p]$ denotes the image in $K_0(A)$ of the class $[p]_{\mathcal{D}} \in \mathcal{D}(A)$. Moreover:

1. For $p, q \in \mathbb{CP}_\infty(A)$ one has $[p] = [q]$ in $K_0(A)$ if and only if there is a projection $r \in \mathbb{CP}_\infty(A)$ with $p \oplus r \sim_0 q \oplus r$.
2. For any two projections $p, q \in \mathbb{CP}_\infty(A)$ the relation $[p \oplus q] = [p] + [q]$ holds in $K_0(A)$.

This is the working description of K_0 that every later computation rests on. The content of part (1) is that equality in $K_0(A)$ is *stable* equivalence: p and q need not be equivalent on the nose, only after adding a common projection r . This is the algebraic analogue of stable triviality for vector bundles ([2]), where two bundles agree in K -theory once a common trivial bundle is added to each. The cancellation that is hidden inside the Grothendieck construction is precisely this padding by r .

Proof:

Write $S = \mathcal{D}(A)$ for the abelian semigroup of definition 1.0.8, with addition induced by $\oplus$. By definition 1.1.1 the group $K_0(A)$ is the Grothendieck group $G(S)$, realized as $(S \times S)/\approx$ where $(x_1, y_1) \approx (x_2, y_2)$ when there is $z \in S$ with $x_1 + y_2 + z = x_2 + y_1 + z$, with the class of (x, y) written $x - y$ and with the canonical semigroup map $\gamma: S \rightarrow G(S)$, $\gamma(x) = x - 0$.

Step 1: Every element is a difference of projection classes. A general element of $G(S)$ is $\gamma(s) - \gamma(t)$ for $s, t \in S$. By definition of $\mathcal{D}(A)$, $s = [p]_{\mathcal{D}}$ and $t = [q]_{\mathcal{D}}$ for projections $p, q \in \mathbb{CP}_\infty(A)$. Writing $[p] := \gamma([p]_{\mathcal{D}})$, the element is $[p] - [q]$, which is the asserted form.

Step 2: Additivity. The map γ is a semigroup homomorphism, so

$$[p \oplus q] = \gamma([p \oplus q]_{\mathcal{D}}) = \gamma([p]_{\mathcal{D}} + [q]_{\mathcal{D}}) = \gamma([p]_{\mathcal{D}}) + \gamma([q]_{\mathcal{D}}) = [p] + [q]$$

using the definition of addition in $\mathcal{D}(A)$ from definition 1.0.8. This proves (2).

Step 3: Equality criterion. By the construction of $G(S)$, one has $\gamma(s) = \gamma(t)$ in $G(S)$ if and only if $(s, 0) \approx (t, 0)$, that is, if and only if there is $z \in S$ with $s + z = t + z$ in S . Taking $s = [p]_{\mathcal{D}}$, $t = [q]_{\mathcal{D}}$ and $z = [r]_{\mathcal{D}}$ for a projection r , the equation $s + z = t + z$ reads $[p \oplus r]_{\mathcal{D}} = [q \oplus r]_{\mathcal{D}}$, which by definition of $\mathcal{D}(A)$ means $p \oplus r \sim_0 q \oplus r$. Conversely any $z \in S$ is of the form $[r]_{\mathcal{D}}$ for some projection r , since $\mathcal{D}(A) = \mathbb{CP}_\infty(A)/\sim_0$. This is exactly statement (1), completing the proof.

A point worth isolating, because it streamlines every later argument, is that on a unital C^* -algebra the von Neumann equivalence $\sim_0$ over $\mathbb{CP}_\infty(A)$ coincides with the coarser-looking notion obtained by allowing conjugation by unitaries in a large enough matrix algebra; both are subsumed by the next standard fact, which lets one trade analytic equivalence for the homotopy of projections.

Lemma 1.1.3: Equivalences of Projections Coincide up to Stabilization

Let A be a unital C^* -algebra and let $p, q \in \mathbb{CP}_n(A)$ be projections in the same matrix algebra. Consider the relations

1. $p \sim_0 q$ (von Neumann: $p = v^*v$, $q = vv^*$ for some $v \in M_n(A)$);
2. $p \sim_h q$ (homotopy: p and q lie in the same path-component of the projections in $M_n(A)$);
3. $p \sim_u q$ (unitary: $q = upu^*$ for some unitary $u \in M_n(A)$).

Then (2) $\Rightarrow$ (3) $\Rightarrow$ (1), and if $\|p - q\| < 1$ then $p \sim_u q$. Conversely, if $p \sim_0 q$ then $p \oplus 0_n \sim_u q \oplus 0_n$ in $M_{2n}(A)$. Consequently all three relations induce the same semigroup $\mathcal{D}(A)$ and the same group $K_0(A)$.

The three notions of when two projections “are the same” are not literally equal as relations, but they generate the same equivalence once one is free to enlarge the ambient matrix size, and K_0 is built precisely to be insensitive to such enlargement. This is the operator-algebraic counterpart of the freedom, in [2], to stabilize a vector bundle by adding trivial summands before comparing.

Proof:

Step 1: (2) $\Rightarrow$ (3). Let $t \mapsto p_t$ be a continuous path of projections in $M_n(A)$ with $p_0 = p$, $p_1 = q$. By a partition of $[0, 1]$ into intervals on which $\|p_s - p_t\| < 1$, it suffices to treat the case $\|p - q\| < 1$. Set

$$z = qp + (1 - q)(1 - p)$$

A direct computation gives $z - 1 = (q - p)(2p - 1)$, so $\|z - 1\| \leq \|q - p\| < 1$ (since $2p - 1$ is a self-adjoint unitary of norm 1), hence z is invertible. Moreover $zp = qp$ and $qz = qp$, so $zp = qz$. Polar-decomposing $z = u|z|$ with $|z| = (z^*z)^{1/2}$ invertible, u is unitary; the relation $zp = qz$ together with its adjoint $pz^* = z^*q$ gives $pz^*z = z^*qz = z^*zp$, so z^*z , and hence $|z|$, commutes with p . Then $upu^* = z|z|^{-1}p|z|^{-1}z^* = zp(z^*z)^{-1}z^* = qz(z^*z)^{-1}z^* = q$, using $z(z^*z)^{-1}z^* = 1$. So $p \sim_u q$.

Step 2: (3) $\Rightarrow$ (1). If $q = upu^*$ with u unitary, put $v = up$. Then $v^*v = pu^*up = p$ and $vv^* = uppu^* = upu^* = q$, so $p \sim_0 q$.

Step 3: (1) $\Rightarrow$ (3) *after stabilizing*. Suppose $p = v^*v$, $q = vv^*$ with $v \in M_n(A)$. Then $v = qvp$, so $vp = v = qv$ and, taking adjoints, $pv^* = v^* = v^*q$. Inside $M_{2n}(A)$ form

$$w = \begin{pmatrix} v & 1 - q \\ 1 - p & v^* \end{pmatrix}$$

A direct computation using $v^*v = p$, $vv^* = q$, $vp = v$, $qv = v$ gives

$$ww^* = \begin{pmatrix} vv^* + (1 - q)^2 & v(1 - p) + (1 - q)v \\ (1 - p)v^* + v^*(1 - q) & (1 - p)^2 + v^*v \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

since $vv^* + (1 - q) = q + 1 - q = 1$, $v(1 - p) + (1 - q)v = (v - v) + (v - v) = 0$, and $(1 - p) + v^*v = 1 - p + p = 1$; the computation $w^*w = 1_{2n}$ is identical. So w is a unitary in $M_{2n}(A)$. Since $vp = v$ and $(1 - p)p = 0$,

$$w(p \oplus 0_n)w^* = \begin{pmatrix} v & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v^* & 1 - p \\ 1 - q & v \end{pmatrix} = \begin{pmatrix} vv^* & v(1 - p) \\ 0 & 0 \end{pmatrix} = q \oplus 0_n$$

using $vv^* = q$ and $v(1 - p) = v - vp = 0$. Hence $p \oplus 0_n \sim_u q \oplus 0_n$.

Since $p \sim_0 p \oplus 0_n$ by proposition 1.0.7(1), the implications (2) $\Rightarrow$ (3) $\Rightarrow$ (1) together with this stabilized converse show that on $\mathbb{CP}_\infty(A)$ all three relations have the same equivalence classes after the harmless padding by zeros. They therefore define the same quotient $\mathcal{D}(A)$ and the same Grothendieck group, as we sought to show.

Example 1.1.4: K_0 of the Complex Numbers

Take $A = \mathbb{C}$, the simplest unital C^* -algebra. A projection in $M_n(\mathbb{C})$ is a self-adjoint idempotent matrix, hence an orthogonal projection onto a subspace of $\mathbb{C}^n$, and is determined up to unitary conjugacy by its rank. Two projections $p \in M_m(\mathbb{C})$, $q \in M_n(\mathbb{C})$ satisfy $p \sim_0 q$ if and only if $\text{rk } p = \text{rk } q$: a partial isometry v with $v^*v = p$, $vv^* = q$ is exactly an isometry from imp onto $\text{im } q$, which exists precisely when the ranks agree. Thus

$$\mathcal{D}(\mathbb{C}) = \mathbb{CP}_\infty(\mathbb{C}) / \sim_0 \xrightarrow{\text{rk}} \mathbb{N}_{\geq 0}$$

is a semigroup isomorphism onto $(\mathbb{N}_{\geq 0}, +)$. The Grothendieck group of $(\mathbb{N}_{\geq 0}, +)$ is $\mathbb{Z}$, with the class of a rank- k projection sent to k . Hence

$$K_0(\mathbb{C}) \cong \mathbb{Z}$$

with the rank-one projection $[1] = [e_{11}]$ as generator. This matches $K^0(\text{pt}) = \mathbb{Z}$ from [2]: a point carries vector bundles classified by their dimension, and $\mathbb{C} = C(\text{pt})$.

With the standard picture in place, the first structural property to establish is functoriality. A $*$ -homomorphism moves projections to projections and respects the equivalence and the block sum, so it descends to the level of K_0 . The substantial part of the statement is homotopy invariance, which is what makes K_0 a genuinely topological invariant rather than a rigid algebraic one.

Theorem 1.1.5: Functoriality and Homotopy Invariance of K_0

The assignment $A \mapsto K_0(A)$ is a covariant functor from unital C^* -algebras and unital $*$ -homomorphisms to abelian groups. Explicitly, a unital $*$ -homomorphism $\varphi: A \rightarrow B$ induces a group homomorphism

$$K_0(\varphi): K_0(A) \rightarrow K_0(B), \quad K_0(\varphi)([p] - [q]) = [\varphi_\infty(p)] - [\varphi_\infty(q)]$$

where $\varphi_\infty: \mathbb{CP}_\infty(A) \rightarrow \mathbb{CP}_\infty(B)$ is the entrywise application of φ , satisfying $K_0(\text{id}) = \text{id}$ and $K_0(\psi \circ \varphi) = K_0(\psi) \circ K_0(\varphi)$. Moreover, if $\varphi_0, \varphi_1: A \rightarrow B$ are homotopic through unital $*$ -homomorphisms (there is a path $t \mapsto \varphi_t$ with each φ_t a unital $*$ -homomorphism and $t \mapsto \varphi_t(a)$ norm-continuous for every $a \in A$), then

$$K_0(\varphi_0) = K_0(\varphi_1)$$

The homotopy-invariance clause is the engine of every computation. It says that K_0 cannot distinguish $*$ -homomorphisms connected by a continuous deformation, so two algebras related by a homotopy equivalence of $*$ -homomorphisms have isomorphic K_0 . It is the operator-algebraic mirror of [2], which states that homotopic maps pull back vector bundles to isomorphic bundles. Although stated here for unital $*$ -homomorphisms, the functor extends to all $*$ -homomorphisms by definition 1.1.8:

an arbitrary $\varphi: A \rightarrow B$ acts on K_0 through its unital extension $\tilde{\varphi}: \tilde{A} \rightarrow \tilde{B}$, and the identities $K_0(\text{id}) = \text{id}$, $K_0(\psi\varphi) = K_0(\psi)K_0(\varphi)$ persist because unitization is itself functorial (proposition 1.0.4). This is what licenses applying K_0 to the non-unital inclusion $\iota: I \rightarrow A$ in theorem 1.1.10.

Proof:

Step 1: Well-definedness and functoriality. A $*$ -homomorphism $\varphi: A \rightarrow B$ induces $\varphi_n: M_n(A) \rightarrow M_n(B)$ entrywise, and since φ preserves adjoints and products it sends a projection $p = p^2 = p^*$ to a projection $\varphi_n(p)$. If $p \sim_0 q$ via v , with $p = v^*v$, $q = vv^*$, then $\varphi(v)$ witnesses $\varphi_n(p) \sim_0 \varphi_n(q)$, because φ is multiplicative and $*$ -preserving. Also φ commutes with $\oplus$ (block diagonal sum), so φ descends to a semigroup homomorphism $\mathcal{D}(A) \rightarrow \mathcal{D}(B)$. By the universal property of the Grothendieck construction (definition 1.1.1) this extends uniquely to the group homomorphism $K_0(\varphi)$ of the stated form. The identities $K_0(\text{id}) = \text{id}$ and $K_0(\psi\varphi) = K_0(\psi)K_0(\varphi)$ are immediate from the corresponding identities for the entrywise maps, since both sides agree on the generating classes $[p]$.

Step 2: Homotopy invariance. Fix a projection $p \in \mathbb{CP}_n(A)$ and a homotopy $t \mapsto \varphi_t$ of unital $*$ -homomorphisms. The map $t \mapsto (\varphi_t)_n(p)$ is a norm-continuous path of projections in $M_n(B)$, since each entry $t \mapsto \varphi_t(p_{ij})$ is continuous by hypothesis and the projection relations $(\varphi_t)_n(p)^2 = (\varphi_t)_n(p) = (\varphi_t)_n(p)^*$ hold for every t . By lemma 1.1.3, a homotopy of projections gives a unitary equivalence, hence $\sim_0$ -equivalence, so

$$[(\varphi_0)_n(p)] = [(\varphi_1)_n(p)] \quad \text{in } K_0(B)$$

Every element of $K_0(A)$ is a difference $[p] - [q]$ of such classes by proposition 1.1.2, and $K_0(\varphi_0), K_0(\varphi_1)$ agree on each, so $K_0(\varphi_0) = K_0(\varphi_1)$, completing the proof.

The next pillar is stability: K_0 does not see the difference between A and its matrix amplifications, nor between A and its stabilization by the compact operators. This is the precise sense in which K_0 is a *stable* invariant, and it is what allows one to compute K_0 of an algebra by replacing it with a more convenient Morita-equivalent partner.

Theorem 1.1.6: Stability of K_0

Let A be a unital C^* -algebra. Then:

1. For every $n \geq 1$, the corner embedding $\iota: A \rightarrow M_n(A)$, $a \mapsto \text{diag}(a, 0, \dots, 0)$, induces an isomorphism

$$K_0(\iota): K_0(A) \xrightarrow{\sim} K_0(M_n(A))$$

2. Let $\mathcal{K} = \mathcal{K}(H)$ be the C^* -algebra of compact operators on a separable infinite-dimensional Hilbert space, and let $e \in \mathcal{K}$ be a rank-one projection. The embedding $A \rightarrow A \otimes \mathcal{K}$, $a \mapsto a \otimes e$, induces an isomorphism

$$K_0(A) \xrightarrow{\sim} K_0(A \otimes \mathcal{K})$$

Part (1) is the statement that “a projection over $M_n(A)$ is just a bigger projection over A ,” formalized through the identification $M_k(M_n(A)) \cong M_{kn}(A)$. Part (2) upgrades this to the limit: $\mathcal{K}$ is the norm closure of the increasing union of the $M_n(\mathbb{C})$, and K_0 is continuous along such limits, so stabilizing by $\mathcal{K}$ changes nothing. Part (2) is also the cleanest statement of *Morita invariance* for K_0 , since A and $A \otimes \mathcal{K}$ are stably isomorphic.

Proof:

Step 1: The matrix isomorphism. There is a natural $*$ -isomorphism $M_k(M_n(A)) \cong M_{kn}(A)$ for every k , regrouping a $k \times k$ matrix of $n \times n$ blocks as a single $kn \times kn$ matrix over A . Under these identifications, the projections over $M_n(A)$ are exactly the projections in $M_{kn}(A)$, and von Neumann equivalence and block sum correspond. Hence the entrywise inclusions assemble to a semigroup isomorphism $\mathcal{D}(A) \rightarrow \mathcal{D}(M_n(A))$: every projection $P \in M_k(M_n(A)) = M_{kn}(A)$ is a projection over A , and conversely. The map on $\mathcal{D}$ induced by ι sends $[p]_{\mathcal{D}}$ (for $p \in M_k(A)$) to the class of $\text{diag}(p, 0, \dots, 0) \in M_k(M_n(A)) = M_{kn}(A)$, which by proposition 1.0.7(1) equals $[p]_{\mathcal{D}}$ under the identification. So the induced map $\mathcal{D}(A) \rightarrow \mathcal{D}(M_n(A))$ is a bijection, hence an isomorphism of semigroups, and passing to Grothendieck groups gives the isomorphism $K_0(\iota)$.

Step 2: Continuity along the union. Write $\mathcal{K} = \overline{\bigcup_n M_n(\mathbb{C})}$, the closure of the union under the corner inclusions $M_n(\mathbb{C}) \hookrightarrow M_{n+1}(\mathbb{C})$, $x \mapsto \text{diag}(x, 0)$. Then $A \otimes \mathcal{K} = \overline{\bigcup_n M_n(A)}$ with the corresponding corner inclusions $j_n: M_n(A) \hookrightarrow M_{n+1}(A)$. Let $P \in \mathcal{CP}_{\infty}(A \otimes \mathcal{K})$, that is, $P \in M_k(A \otimes \mathcal{K})$ a projection. Since $\bigcup_n M_k(M_n(A))$ is dense in $M_k(A \otimes \mathcal{K})$, choose x in the union with $\|x - P\| < 1/4$; replacing x by $(x + x^*)/2$ we may take x self-adjoint with $\|x - P\| < 1/2$. The spectrum of x then misses $\{1/2\}$, and the spectral projection $P' = \mathbb{1}_{(1/2, \infty)}(x)$, computed by holomorphic functional calculus, is a projection lying in some $M_k(M_n(A))$ with $\|P' - P\| < 1$, whence $P' \sim_0 P$ by lemma 1.1.3. So every class in $\mathcal{D}(A \otimes \mathcal{K})$ comes from some $\mathcal{D}(M_n(A))$, and the colimit of the system

$$\mathcal{D}(M_1(A)) \rightarrow \mathcal{D}(M_2(A)) \rightarrow \dots$$

maps onto $\mathcal{D}(A \otimes \mathcal{K})$. Injectivity at the level of $\mathcal{D}$ holds because if two projections in some $M_n(A)$ become $\sim_0$ in $A \otimes \mathcal{K}$, the implementing v can be approximated within the union, and a small perturbation of a partial isometry between projections is again such (by polar decomposition, as in lemma 1.1.3); thus they are already $\sim_0$ in some $M_m(A)$. Hence

$$\mathcal{D}(A \otimes \mathcal{K}) \cong \varinjlim_n \mathcal{D}(M_n(A))$$

By Step 1 each $\mathcal{D}(M_n(A)) \cong \mathcal{D}(A)$ compatibly with the corner inclusions, which are the identity on $\mathcal{D}(A)$, so the colimit is $\mathcal{D}(A)$ itself. Passing to Grothendieck groups, $K_0(A \otimes \mathcal{K}) \cong K_0(A)$, and tracing the isomorphism shows it is induced by $a \mapsto a \otimes e$, as we sought to show.

Example 1.1.7: K_0 of Matrix Algebras and of the Compacts

Applying theorem 1.1.6(1) with $A = \mathbb{C}$ gives

$$K_0(M_n(\mathbb{C})) \cong K_0(\mathbb{C}) \cong \mathbb{Z}$$

for every n . The generator is the class of a rank-one projection $e_{11} \in M_n(\mathbb{C})$; under the isomorphism $K_0(M_n(\mathbb{C})) \cong \mathbb{Z}$ a projection of rank r in $M_n(\mathbb{C})$ (equivalently a projection in some $M_k(M_n(\mathbb{C})) = M_{kn}(\mathbb{C})$ of rank r) maps to r . The corner map $\mathbb{C} \rightarrow M_n(\mathbb{C})$ sends the generator $1 \mapsto e_{11}$, so on K_0 it is the identity $\mathbb{Z} \rightarrow \mathbb{Z}$, *not* multiplication by n : K_0 measures rank, which the corner embedding preserves.

Applying theorem 1.1.6(2) with $A = \mathbb{C}$ gives

$$K_0(\mathcal{K}) \cong K_0(\mathbb{C}) \cong \mathbb{Z}$$

Concretely, $\mathcal{K} = \mathbb{C} \otimes \mathcal{K}$ and every projection in $\mathcal{K}$ is a finite-rank orthogonal projection, of some rank $r \in \mathbb{N}_{\geq 0}$; the rank is a complete invariant for $\sim_0$, so $\mathcal{D}(\mathcal{K}) \cong (\mathbb{N}_{\geq 0}, +)$ and $K_0(\mathcal{K}) \cong \mathbb{Z}$ with

a rank-one projection as generator. The unital algebra $B(H)$ behaves entirely differently: there $1 \sim_0 1 \oplus 1$ by an isometry $H \cong H \oplus H$, forcing $[1] = 2[1]$, so $K_0(B(H)) = 0$.

So far everything has been unital. The definition definition 1.1.1 requires a unit because the semigroup $\mathcal{D}(A)$ is built from projections, and a non-unital algebra may have very few of them (for instance $C_0(\mathbb{R})$ has no nonzero projections at all). The fix is to pass to the unitization $\tilde{A}$ of proposition 1.0.4, which always has a unit, and then to cut out the contribution of the adjoined scalars. The unitization sits in a split exact sequence

$$0 \longrightarrow A \longrightarrow \tilde{A} \xrightarrow{\pi} \mathbb{C} \longrightarrow 0$$

where $\pi(a, \alpha) = \alpha$ is the quotient map and the splitting $\mathbb{C} \rightarrow \tilde{A}$, $\alpha \mapsto (0, \alpha)$, is a unital $*$ -homomorphism. Applying the functor K_0 to π and the splitting gives a retraction of $K_0(\tilde{A})$ onto $K_0(\mathbb{C}) = \mathbb{Z}$, and the part it retracts away is the right definition of $K_0(A)$.

Definition 1.1.8: K_0 of a Non-Unital C^* -Algebra

Let A be a (not necessarily unital) C^* -algebra with unitization $\tilde{A}$ (proposition 1.0.4) and scalar quotient $\pi: \tilde{A} \rightarrow \mathbb{C}$. Define

$$K_0(A) := \ker(K_0(\pi): K_0(\tilde{A}) \rightarrow K_0(\mathbb{C}) = \mathbb{Z})$$

For a $*$ -homomorphism $\varphi: A \rightarrow B$, the unital extension $\tilde{\varphi}: \tilde{A} \rightarrow \tilde{B}$ commutes with the scalar quotients, so $K_0(\tilde{\varphi})$ restricts to the kernels and defines $K_0(\varphi): K_0(A) \rightarrow K_0(B)$.

When A is already unital this definition agrees with definition 1.1.1: in that case $\tilde{A} \cong A \oplus \mathbb{C}$ as C^* -algebras (proposition 1.0.4), the quotient π is projection onto the second factor, and since a projection over $A \oplus \mathbb{C}$ is a pair of projections (one over each factor), the semigroup $\mathcal{D}(A \oplus \mathbb{C}) \cong \mathcal{D}(A) \times \mathcal{D}(\mathbb{C})$ factors, giving $K_0(\tilde{A}) \cong K_0(A) \oplus \mathbb{Z}$; the kernel of $K_0(\pi)$ is then exactly the original $K_0(A)$. So definition 1.1.8 extends the unital definition without conflict, and from now on K_0 is defined on all C^* -algebras. In general (unital or not) the unital scalar inclusion $\lambda: \mathbb{C} \rightarrow \tilde{A}$, $\alpha \mapsto (0, \alpha)$, satisfies $\pi \circ \lambda = \text{id}_{\mathbb{C}}$, so $K_0(\pi)$ is a split surjection onto $\mathbb{Z}$ with section $K_0(\lambda)$, and therefore

$$K_0(\tilde{A}) \cong \ker K_0(\pi) \oplus \text{im} K_0(\lambda) = K_0(A) \oplus \mathbb{Z}$$

with the $\mathbb{Z}$ summand detecting the rank of the scalar part. This elementary splitting, depending only on the unital section λ , is what the reduction in theorem 1.1.10 uses.

Example 1.1.9: K_0 of $C_0(\mathbb{R})$ via Unitization

Let $A = C_0(\mathbb{R})$, the continuous functions on $\mathbb{R}$ vanishing at infinity, a non-unital algebra with no nonzero projections (a projection in a commutative C^* -algebra is the indicator of a clopen set, and $\mathbb{R}$ is connected with the only clopen sets $\emptyset, \mathbb{R}$, the latter excluded by vanishing at infinity). Its unitization is $\tilde{A} = C(S^1)$, the one-point compactification adding the point at infinity, with π evaluation at that basepoint. Anticipating theorem 1.1.11, $K_0(C(S^1)) \cong K^0(S^1) \cong \mathbb{Z}$, generated by the trivial line bundle, and $K_0(\pi)$ sends this generator to the rank, an isomorphism onto $\mathbb{Z}$. Hence

$$K_0(C_0(\mathbb{R})) = \ker(K_0(C(S^1)) \rightarrow \mathbb{Z}) = 0$$

matching the reduced group $\tilde{K}^0(S^1) = 0$ from [2]: every complex vector bundle over S^1 is trivial.

The reason the unitization-based definition is the correct one is that it makes K_0 behave well on short exact sequences. A $*$ -homomorphism that is injective with closed image, followed by the quotient, forms a short exact sequence of C^* -algebras; the content of the next theorem is that K_0 takes such a sequence to an exact sequence of abelian groups, at least in the middle term. Full exactness on the two ends fails in general (that failure is exactly what the index map and the six-term sequence of section 1.3 will repair), but middle exactness, called half-exactness, already holds and is what powers Mayer–Vietoris-style computations.

Theorem 1.1.10: Half-Exactness of K_0

Let

$$0 \longrightarrow I \xrightarrow{\iota} A \xrightarrow{q} A/I \longrightarrow 0$$

be a short exact sequence of C^* -algebras, where I is a closed two-sided ideal of A , ι the inclusion, and q the quotient. Then the induced sequence

$$K_0(I) \xrightarrow{K_0(\iota)} K_0(A) \xrightarrow{K_0(q)} K_0(A/I)$$

is exact at $K_0(A)$, that is, $\text{im} K_0(\iota) = \ker K_0(q)$. If moreover the sequence splits by a $*$ -homomorphism $s: A/I \rightarrow A$ with $q \circ s = \text{id}$, then

$$0 \longrightarrow K_0(I) \xrightarrow{K_0(\iota)} K_0(A) \xrightarrow{K_0(q)} K_0(A/I) \longrightarrow 0$$

is split exact.

Half-exactness is what makes K_0 a K -theory rather than just a functor. It is the algebraic shadow of the long exact sequence of a pair in topological K -theory ([2]): the ideal I plays the role of the open complement, A/I the role of the closed subspace, and the sequence records how projections (bundles) on the whole space relate to those on the pieces. The split case is what underlies the computation of K_0 of unitizations and of products.

Proof:

Reduction to unital algebras. Unitizing the sequence gives

$$0 \longrightarrow I \longrightarrow \tilde{A} \xrightarrow{\tilde{q}} \widetilde{A/I} \longrightarrow 0$$

where I remains a closed ideal (now of the unital algebra $\tilde{A}$), and $\tilde{A}/I \cong \widetilde{A/I}$ since $\tilde{A}$ adjoins one scalar unit, which descends to the unit of $\widetilde{A/I}$. By definition 1.1.8 the splittings give compatible decompositions $K_0(\tilde{A}) \cong K_0(A) \oplus \mathbb{Z}$ and $K_0(\widetilde{A/I}) \cong K_0(A/I) \oplus \mathbb{Z}$, under which $K_0(\tilde{q})$ is $K_0(q) \oplus \text{id}_{\mathbb{Z}}$ and the maps to and from the $\mathbb{Z}$ -summands are identities. Therefore exactness of $K_0(I) \rightarrow K_0(\tilde{A}) \rightarrow K_0(\widetilde{A/I})$ at the middle is equivalent to exactness of $K_0(I) \rightarrow K_0(A) \rightarrow K_0(A/I)$ at the middle. So it suffices to prove the theorem when A (hence the quotient) is unital and the maps are unital, which is assumed henceforth; note I itself need not be unital, and $K_0(I)$ is understood via definition 1.1.8.

Step 1: $\text{im} K_0(\iota) \subseteq \ker K_0(q)$. Since $q \circ \iota = 0$ as a map (the composite $I \rightarrow A \rightarrow A/I$ is zero), functoriality (theorem 1.1.5) gives $K_0(q) \circ K_0(\iota) = K_0(q \circ \iota) = 0$. Hence the image of $K_0(\iota)$ lies in the kernel of $K_0(q)$.

Step 2: $\ker K_0(q) \subseteq \text{im} K_0(\iota)$. Let $x \in \ker K_0(q)$. By proposition 1.1.2 write $x = [p] - [q_0]$

with $p, q_0 \in \mathbb{CP}_n(A)$ projections; adding the complement $1_n - q_0$ to both terms, we may take $q_0 = 1_m \oplus 0$ a trivial projection, so $x = [p] - [1_m \oplus 0]$ with $p \in \mathbb{CP}_n(A)$. The hypothesis $K_0(q)(x) = 0$ says $[q_n(p)] = [1_m \oplus 0]$ in $K_0(A/I)$, where $q_n: M_n(A) \rightarrow M_n(A/I)$ is the entrywise quotient. By proposition 1.1.2(1), after enlarging n and adding a common trivial projection there is a unitary $\bar{u} \in M_n(A/I)$ with $\bar{u} q_n(p) \bar{u}^* = 1_m \oplus 0$.

Lifting the unitary. Replacing $\bar{u}$ by $\text{diag}(\bar{u}, \bar{u}^*) \in M_{2n}(A/I)$ and the projection $1_m \oplus 0$ by $(1_m \oplus 0) \oplus 0_n$, we may assume $\bar{u}$ lies in the identity component of the unitary group. Indeed $\text{diag}(\bar{u}, \bar{u}^*)$ is connected to 1_{2n} by the explicit Whitehead path

$$t \mapsto \begin{pmatrix} \bar{u} & 0 \\ 0 & 1 \end{pmatrix} R_t \begin{pmatrix} \bar{u}^* & 0 \\ 0 & 1 \end{pmatrix} R_t^{-1}, \quad R_t = \begin{pmatrix} \cos(\frac{\pi t}{2}) & -\sin(\frac{\pi t}{2}) \\ \sin(\frac{\pi t}{2}) & \cos(\frac{\pi t}{2}) \end{pmatrix}$$

which equals 1_{2n} at $t = 0$ and $\text{diag}(\bar{u}, \bar{u}^*)$ at $t = 1$. The identity component of the unitary group of a unital C^* -algebra consists of finite products of exponentials e^{ih} with $h = h^*$, and each such factor lifts: choosing a self-adjoint preimage $h' \in M_n(A)$ of h under the surjection q_n (available since q_n is a surjective $*$ -homomorphism and $(c + c^*)/2$ lifts any self-adjoint h from a preimage c), the element $e^{ih'}$ is a unitary in $M_n(A)$ with $q_n(e^{ih'}) = e^{ih}$. Multiplying these lifts yields a unitary $u \in M_n(A)$ with $q_n(u) = \bar{u}$. Replacing p by upu^* (which lies in the same $\sim_0$ -class, hence gives the same $[p]$) we may assume $q_n(p) = 1_m \oplus 0$ exactly.

Producing a class in $K_0(I)$. Now $p - (1_m \oplus 0) \in M_n(I)$, since q_n annihilates it. Both p and $1_m \oplus 0$ are projections in $M_n(\tilde{I})$ (the matrices over the unitization $\tilde{I}$ of I), each mapping to $1_m \oplus 0$ under the scalar quotient $M_n(\tilde{I}) \rightarrow M_n(\mathbb{C})$. Hence the difference

$$[p] - [1_m \oplus 0] \in \ker(K_0(\tilde{I}) \rightarrow \mathbb{Z}) = K_0(I)$$

has vanishing scalar part and so defines a class in $K_0(I)$ by definition 1.1.8. Its image under $K_0(\iota)$ is $[p] - [1_m \oplus 0] = x$ in $K_0(A)$. Therefore $x \in \text{im} K_0(\iota)$, proving exactness at $K_0(A)$.

Step 3: The split case. Suppose $s: A/I \rightarrow A$ is a $*$ -homomorphism with $qs = \text{id}$. Then $K_0(q)K_0(s) = K_0(qs) = \text{id}$ by theorem 1.1.5, so $K_0(q)$ is surjective with section $K_0(s)$. It remains only to prove that $K_0(\iota)$ is injective; granting this, Step 2 shows $K_0(\iota)$ maps $K_0(I)$ isomorphically onto $\ker K_0(q)$, and combining with the section gives, for any $\xi \in K_0(A)$, the decomposition $\xi = (\xi - K_0(s)K_0(q)(\xi)) + K_0(s)K_0(q)(\xi)$ with first summand in $\ker K_0(q) = \text{im} K_0(\iota)$ and $\text{im} K_0(s) \cap \ker K_0(q) = 0$ (since $K_0(q)$ restricted to $\text{im} K_0(s)$ is injective, its composite with $K_0(s)$ being the identity), so that

$$K_0(A) = \text{im} K_0(\iota) \oplus \text{im} K_0(s) \cong K_0(I) \oplus K_0(A/I)$$

Injectivity of $K_0(\iota)$. The content here is that the section forces injectivity even though no $*$ -homomorphism retraction $A \rightarrow I$ exists in general (the failure of such a retraction is exactly what the index map of section 1.3 measures), so this is not a formal consequence of the tools of Steps 1–2 alone. The clean argument runs through the suspension $SA = C_0(\mathbb{R}) \otimes A$ and the mapping cone of s , machinery built only in section 1.3: a section s splits the inclusion $I \hookrightarrow A$ up to homotopy after suspending, and the homotopy invariance of K_0 (theorem 1.1.5) then converts that homotopy splitting into an honest left inverse $K_0(A) \rightarrow K_0(I)$ for $K_0(\iota)$. Because the suspension and mapping-cone apparatus is not yet available at this point in the development, the verification is deferred: the statement that a homotopy-invariant, half-exact functor on C^* -algebras carries split short exact sequences to split short exact sequences (so that $K_0(\iota)$ is injective whenever a $*$ -homomorphism section is present) is proved in full for K_0 in [6, Chapter 4,

“The functor K_0 ”], and as the general axiomatic implication “half-exact and homotopy invariant $\Rightarrow$ split exact” in [1]. Theorems 1.1.5 and 1.1.6 supply the homotopy invariance and stability hypotheses, and Steps 1–2 supply half-exactness, so those references apply verbatim to the present K_0 .

With injectivity of $K_0(\iota)$ in hand, the sequence

$$0 \longrightarrow K_0(I) \xrightarrow{K_0(\iota)} K_0(A) \xrightarrow{K_0(q)} K_0(A/I) \longrightarrow 0$$

is split exact, completing the proof.

The half-exactness theorem isolates the structural fact; the bridge to topology turns it into geometry. The link is the Serre–Swan correspondence, which identifies the projections over $C(X)$ with the vector bundles over X . [2] already records the geometric half of this: a vector bundle over a compact Hausdorff space is a direct summand of a trivial bundle, hence is “projective” in the bundle category. The algebraic half is that such complemented summands of $C(X)^n$ are exactly the images of projection matrices over $C(X)$. Putting the two together identifies $K_0(C(X))$ with the topological $K^0(X)$ of [2].

Theorem 1.1.11: Serre–Swan: $K_0(C(X)) \cong K^0(X)$

Let X be a compact Hausdorff space and $C(X)$ the unital C^* -algebra of continuous complex-valued functions on X . There is an isomorphism of abelian groups

$$K_0(C(X)) \cong K^0(X)$$

natural in X , under which the class $[p]$ of a projection $p \in M_n(C(X))$ corresponds to the class of the vector bundle $E_p \rightarrow X$ whose fiber at x is the image $p(x)(\mathbb{C}^n) \subseteq \mathbb{C}^n$. Under this isomorphism the rank homomorphism $K_0(C(X)) \rightarrow \mathbb{Z}$ corresponds to the dimension of the bundle, so the kernel $\tilde{K}^0(X)$ corresponds to $K_0(C_0(X \setminus \{x_0\}))$ for any basepoint x_0 .

This is the keystone of the whole subject: it says that the operator K_0 and the topological K^0 are literally the same group, computed two ways. Geometrically one classifies vector bundles; algebraically one classifies projections in matrix algebras over the function algebra. The Gelfand–Naimark duality (theorem 1.0.3) makes $X \mapsto C(X)$ a faithful (contravariant) embedding of compact Hausdorff spaces into C^* -algebras, and Serre–Swan promotes that embedding to an identity of K -theories.

Proof:

Step 1: From projections to bundles. Let $p \in M_n(C(X))$ be a projection, viewed as a continuous map $p: X \rightarrow M_n(\mathbb{C})$ with $p(x) = p(x)^2 = p(x)^*$ for all x , that is, a continuous field of orthogonal projections in $\mathbb{C}^n$. The ranks $\text{rk} p(x)$ form a locally constant, hence (on each component) constant, function of x , because nearby projections of equal norm-distance less than 1 have equal rank. Define

$$E_p = \{(x, v) \in X \times \mathbb{C}^n : v \in p(x)\mathbb{C}^n\}$$

with projection to X . This is a subbundle of the trivial bundle $X \times \mathbb{C}^n$: locally, the continuity of p and the constancy of the rank let one choose, near each x_0 , continuous sections spanning $p(x)\mathbb{C}^n$ (apply the Gram–Schmidt process of [2] to continuous lifts of a basis of $p(x_0)\mathbb{C}^n$), giving a local trivialization. So E_p is a complex vector bundle over X , a direct summand of the trivial bundle with complement E_{1-p} .

Step 2: From bundles to projections. Conversely, let $E \rightarrow X$ be a complex vector bundle. By [2], compactness gives a bundle E' with $E \oplus E' \cong X \times \mathbb{C}^n$ trivial for some n . The orthogonal projection of $\mathbb{C}^n$ onto the E -summand, fiberwise, is a continuous family $p_E: X \rightarrow M_n(\mathbb{C})$ of projections (continuous because, after choosing a Hermitian metric via [2], the orthogonal projection onto a continuously varying subspace varies continuously), hence $p_E \in M_n(C(X))$ is a projection with $E_{p_E} \cong E$. So every bundle arises as some E_p .

Step 3: The correspondence respects equivalence and sum. If $p \sim_0 q$ over $C(X)$, the implementing partial isometry v is a continuous field of partial isometries, giving a fiberwise isometric isomorphism $E_p \cong E_q$; conversely an isomorphism $E_p \cong E_q$ of subbundles of trivial bundles is implemented by such a v after stabilizing, by lemma 1.1.3. Block sum of projections corresponds to direct sum of bundles: $E_{p \oplus q} \cong E_p \oplus E_q$ by inspection of fibers. Therefore the assignment $p \mapsto E_p$ descends to a semigroup isomorphism

$$\mathcal{D}(C(X)) \xrightarrow{\sim} (\mathbb{V}\mathbb{B}_{\mathbb{C}}(X), \oplus)$$

between the projection semigroup of definition 1.0.8 and the monoid of isomorphism classes of complex vector bundles under Whitney sum ([2]).

Step 4: Passing to Grothendieck groups. The group $K_0(C(X))$ is the Grothendieck group of $\mathcal{D}(C(X))$ by definition 1.1.1, and $K^0(X) = KU(X)$ is the Grothendieck group of $(\mathbb{V}\mathbb{B}_{\mathbb{C}}(X), \oplus)$ by [2]. A semigroup isomorphism induces an isomorphism of Grothendieck groups, so

$$K_0(C(X)) \cong K^0(X)$$

Naturality in X holds because a continuous map $f: X \rightarrow Y$ induces the unital $*$ -homomorphism $f^*: C(Y) \rightarrow C(X)$ on one side and the bundle pullback f^* on the other, and $E_{f^*(p)} = f^*(E_p)$ by definition of pullback. The rank of $p \in M_n(C(X))$ is the dimension of E_p , so the rank homomorphism matches the dimension homomorphism $K^0(X) \rightarrow \mathbb{Z}$ of [2]; its kernel is the reduced group $\tilde{K}^0(X)$. Identifying $\tilde{K}^0(X)$ with $K_0(C_0(X \setminus \{x_0\}))$ follows from definition 1.1.8 applied to the split sequence $0 \rightarrow C_0(X \setminus \{x_0\}) \rightarrow C(X) \rightarrow \mathbb{C} \rightarrow 0$ (evaluation at x_0), whose K_0 is split exact by theorem 1.1.10, giving $K_0(C(X)) \cong K_0(C_0(X \setminus \{x_0\})) \oplus \mathbb{Z}$ with the $\mathbb{Z}$ the rank. This completes the proof.

The Serre–Swan identification turns every computation of topological K^0 into a computation of operator K_0 and conversely. The flagship example is the 2-sphere, where the answer $\mathbb{Z}^2$ encodes both the rank of a bundle and the first Chern number, and where the operator side produces the same $\mathbb{Z}^2$ through a single nontrivial projection, the Bott projection.

Example 1.1.12: $K_0(C(S^2)) \cong \mathbb{Z}^2$ and the Bott Projection

From theorem 1.1.11, $K_0(C(S^2)) \cong K^0(S^2)$. By [2] the reduced group splits off a free summand,

$$K^0(S^2) \cong \tilde{K}^0(S^2) \oplus \mathbb{Z}$$

and $\tilde{K}^0(S^2) \cong \mathbb{Z}$ because, by [2], complex n -plane bundles over S^2 are classified by homotopy classes of clutching functions $S^1 \rightarrow \mathrm{GL}_n(\mathbb{C})$, that is, by $\pi_1(\mathrm{GL}_n(\mathbb{C})) \cong \pi_1(U(n)) \cong \mathbb{Z}$ (the last isomorphism is the determinant $U(n) \rightarrow U(1) = S^1$); these stabilize to the single group $\tilde{K}^0(S^2) \cong \mathbb{Z}$, with generator the class of the canonical line bundle H over $\mathbb{CP}^1 = S^2$ (clutching function $z \mapsto z$). Hence

$$K_0(C(S^2)) \cong \mathbb{Z}^2$$

with basis given by the class $[1]$ of the trivial line bundle (rank) and the class $[H] - [1]$ of the reduced generator (first Chern number).

On the operator side this $\mathbb{Z}^2$ is produced explicitly by the *Bott projection*. Write $S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$ and set

$$p_B = \frac{1}{2} \begin{pmatrix} 1 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & 1 - x_3 \end{pmatrix} \in M_2(C(S^2))$$

A direct check using $x_1^2 + x_2^2 + x_3^2 = 1$ gives $p_B = p_B^2 = p_B^*$, so p_B is a projection of constant rank 1 (its trace is 1). The line bundle E_{p_B} has first Chern number ± 1 (it is the canonical line bundle H , up to the orientation of S^2), so it generates $\tilde{K}^0(S^2) \cong \mathbb{Z}$, and

$$[p_B] - [e_{11}] \in K_0(C(S^2))$$

is a reduced generator, agreeing with $\pm([H] - [1])$. Together with $[e_{11}]$ (rank) it gives a basis of $\mathbb{Z}^2$. The matching with the topological computation is the content of theorem 1.1.11: the operator-algebraic projection p_B and the geometric clutching function $z \mapsto z$ describe the same generator of $\tilde{K}^0(S^2) \cong \mathbb{Z}$.

Exercise 1.1.1

1. Let A be a unital C^* -algebra and $p, q \in \mathbb{CP}_n(A)$ projections with $\|p - q\| < 1$. Show directly that $p \sim_u q$, and deduce that $[p] = [q]$ in $K_0(A)$. (This is the local step behind homotopy invariance; reconstruct it without quoting lemma 1.1.3.)
2. Compute $K_0(\mathbb{C} \oplus \mathbb{C})$ and, more generally, $K_0(\bigoplus_{i=1}^k M_{n_i}(\mathbb{C}))$ for a finite-dimensional C^* -algebra. Identify the rank homomorphism in each summand and the class of the unit.
3. Using definition 1.1.8 and the split sequence $0 \rightarrow C_0(\mathbb{R}^2) \rightarrow C(S^2) \rightarrow \mathbb{C} \rightarrow 0$ (evaluation at a basepoint), compute $K_0(C_0(\mathbb{R}^2))$. Reconcile your answer with $\tilde{K}^0(S^2)$ from example 1.1.12.
4. Verify that the Bott projection $p_B \in M_2(C(S^2))$ of example 1.1.12 satisfies $p_B = p_B^2 = p_B^*$ and has constant rank 1. Then show that p_B is *not* Murray–von Neumann equivalent to a constant rank-one projection e_{11} , and explain what invariant obstructs the equivalence.
5. Let X be a compact Hausdorff space with $X = X_1 \sqcup X_2$ a disjoint union of two clopen pieces. Using theorem 1.1.11, prove $K_0(C(X)) \cong K_0(C(X_1)) \oplus K_0(C(X_2))$ both topologically (via K^0) and algebraically (via the C^* -algebra decomposition $C(X) \cong C(X_1) \oplus C(X_2)$).
6. Show that $K_0(B(H)) = 0$ for H a separable infinite-dimensional Hilbert space, and contrast this with $K_0(\mathcal{K}(H)) \cong \mathbb{Z}$ from example 1.1.7. Which hypothesis in theorem 1.1.6 fails for $B(H)$, and why does the argument for $\mathcal{K}$ not apply?

1.2 K_1 of a C^* -Algebra

The group $K_0(A)$ of definition 1.1.1 records projections, the “zero-dimensional” data of a C^* -algebra, by mimicking the Grothendieck construction for vector bundles. The natural companion invariant records the unitaries: where K_0 assembles idempotent self-adjoint elements into a group, K_1 assembles

the symmetries (unitaries) of the matrix algebras $M_n(A)$, modulo the relation of being connected by a continuous path. This section defines $K_1(A)$ for a unital C^* -algebra A as the set of path components of the infinite unitary group, proves that it is an abelian group, establishes its functoriality, homotopy invariance, and stability, and identifies $K_1(C(X))$ with the topological group $K^1(X)$ for compact X . The construction is governed throughout by the analytic structure of a C^* -algebra (the norm topology, the polar decomposition, the continuous functional calculus), and it is precisely this analytic input that distinguishes the operator-theoretic K_1 from the purely algebraic Whitehead group $\mathrm{GL}(R)/E(R)$ of [3]. The distinction is drawn carefully at the end of the section.

Throughout, A is a unital C^* -algebra (recall definition 1.0.1) unless stated otherwise; the non-unital case is reduced to the unital one by passing to the unitization $\tilde{A}$ of proposition 1.0.4 at the very end. For a unital C^* -algebra A write

$$\mathcal{U}_n(A) = \{u \in M_n(A) \mid u^*u = uu^* = 1_n\}$$

for the unitary group of the matrix algebra $M_n(A)$, where 1_n is the identity of $M_n(A)$. The block-diagonal inclusion

$$\mathcal{U}_n(A) \hookrightarrow \mathcal{U}_{n+1}(A) \quad u \mapsto \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix}$$

is a continuous group homomorphism, and the colimit (union) over these inclusions is the infinite unitary group

$$\mathcal{U}_\infty(A) = \varinjlim_n \mathcal{U}_n(A) = \bigcup_{n=1}^{\infty} \mathcal{U}_n(A)$$

where $u \in \mathcal{U}_n(A)$ is identified with its image $u \oplus 1$ in $\mathcal{U}_{n+1}(A)$. The group $\mathcal{U}_\infty(A)$ carries the colimit topology: a set is open precisely when its intersection with each $\mathcal{U}_n(A)$ is open in the norm topology. The two operations available on $\mathcal{U}_\infty(A)$ are the group multiplication inherited from each $\mathcal{U}_n(A)$ and the block sum $u \oplus v = \mathrm{diag}(u, v)$; the interplay between them is the engine of the whole theory.

Before stating the definition, the reason for taking path components rather than the unitary group itself deserves a word. A single unitary $u \in \mathcal{U}_n(A)$ carries spectral information that is too fine to be a homotopy invariant: the unitaries connected to the identity through a continuous path form a normal subgroup, and quotienting by it discards exactly the data that varies continuously, retaining the discrete obstruction to contractibility. This obstruction is what K_1 measures.

Definition 1.2.1: K_1 of a C^* -Algebra

Let A be a unital C^* -algebra. The K_1 -group of A is the set of path components of the infinite unitary group,

$$K_1(A) = \pi_0(\mathcal{U}_\infty(A)) = \mathcal{U}_\infty(A)/\mathcal{U}_\infty(A)_0$$

where $\mathcal{U}_\infty(A)_0$ is the path component of the identity. The class of a unitary $u \in \mathcal{U}_n(A)$ in $K_1(A)$ is written $[u]_1$, and the group operation is induced by the block sum,

$$[u]_1 + [v]_1 = [u \oplus v]_1 = \left[\begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix} \right]_1$$

with identity element $[1]_1 = 0$.

The definition asserts more than it has so far justified: that π_0 of $\mathcal{U}_\infty(A)$ is a group at all, that $\mathcal{U}_\infty(A)_0$ is a normal subgroup so that the quotient makes sense, that the operation is well-defined on

path components, and that the resulting group is abelian. All of these follow from a single mechanism, the matrix manoeuvre known as the Whitehead lemma, which says that in the infinite unitary group the block sum $u \oplus v$ is path-connected to the ordinary product $uv \oplus 1$. The verification occupies theorem 1.2.4 below. For now, the intuition is that $K_1(A)$ measures how many “essentially different” invertible symmetries the matrix algebras of A support: an element is trivial in K_1 exactly when, after enlarging the matrix size, it can be continuously deformed to the identity.

Two preliminary reductions make $\mathcal{U}_\infty(A)$ tractable and tie the definition to the invertible (rather than unitary) elements. The first is that, in a unital C^* -algebra, the unitary group of any matrix algebra is path-connected to the identity exactly when the corresponding general linear group is, because the polar decomposition retracts invertibles onto unitaries continuously. The second is the continuous logarithm available from the functional calculus. Both are recorded now.

Lemma 1.2.2: Connectivity via the Exponential

Let A be a unital C^* -algebra and $u \in \mathcal{U}_n(A)$. If $u = e^{ih}$ for some self-adjoint $h \in M_n(A)$, then the path $t \mapsto e^{it h}$, $t \in [0, 1]$, is a continuous path of unitaries in $\mathcal{U}_n(A)$ from 1 to u . Conversely, every $u \in \mathcal{U}_n(A)$ with $\|u - 1\| < 2$ is of this form, and in particular lies in $\mathcal{U}_n(A)_0$.

Proof:

Step 1: Exponentials give paths. Suppose $u = e^{ih}$ with $h = h^* \in M_n(A)$. For each $t \in [0, 1]$ the element th is self-adjoint, so $e^{it h}$ is unitary: indeed $(e^{it h})^* = e^{-it h}$ by continuity of the involution applied to the convergent exponential series, and $e^{it h} e^{-it h} = 1$ since th commutes with itself. The map $t \mapsto e^{it h} = \sum_{k \geq 0} (it h)^k / k!$ is norm-continuous in t because the series converges uniformly for t in the compact interval $[0, 1]$ (the k -th term is bounded by $\|h\|^k / k!$). At $t = 0$ it is 1 and at $t = 1$ it is u , giving the asserted path inside $\mathcal{U}_n(A)$.

Step 2: Nearby unitaries are exponentials. Suppose $u \in \mathcal{U}_n(A)$ satisfies $\|u - 1\| < 2$. The spectrum $\sigma(u)$ lies in the unit circle $\mathbb{T}$ (as u is unitary) and avoids the point -1 , because $-1 \in \sigma(u)$ would force $\|u - 1\| \geq |-1 - 1| = 2$ by the spectral radius identity $\|u - 1\| = \sup \{|\lambda - 1| \mid \lambda \in \sigma(u)\}$ valid for the normal element $u - 1$. Thus $\sigma(u)$ is contained in the open arc $\mathbb{T} \setminus \{-1\}$, on which the principal branch of the logarithm $\log : \mathbb{T} \setminus \{-1\} \rightarrow (-i\pi, i\pi)$, $e^{i\theta} \mapsto i\theta$ for $\theta \in (-\pi, \pi)$, is continuous. Applying the continuous functional calculus to the normal element u yields $h := -i \log(u) \in M_n(A)$, which is self-adjoint because $\log$ maps $\mathbb{T} \setminus \{-1\}$ into $i\mathbb{R}$ so that $-i \log$ is real-valued on $\sigma(u)$, and which satisfies $e^{ih} = u$ by the functional-calculus identity $\exp(i \cdot (-i \log)) = \text{id}$ on $\sigma(u)$. By Step 1, $u \in \mathcal{U}_n(A)_0$, completing the proof.

The exponential path is the basic building block: it shows that the identity component $\mathcal{U}_n(A)_0$ contains every unitary within distance 2 of the identity, hence (being a subgroup closed under the path-connecting operation) is open, and that $\mathcal{U}_n(A)_0$ is generated by exponentials of self-adjoint elements. The same lemma applied in M_∞ shows $\mathcal{U}_\infty(A)_0$ is generated by such exponentials. The next lemma is the decisive algebraic identity.

Lemma 1.2.3: Whitehead Lemma for Unitaries

Let A be a unital C^* -algebra and let $u, v \in \mathcal{U}_n(A)$. Then in $\mathcal{U}_{2n}(A)$ the following hold.

1. The block unitary $\begin{pmatrix} u & 0 \\ 0 & u^* \end{pmatrix}$ lies in $\mathcal{U}_{2n}(A)_0$; it is connected to 1_{2n} by an explicit path of unitaries.
2. The block unitaries $\begin{pmatrix} uv & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix}$ are connected by a path of unitaries in $\mathcal{U}_{2n}(A)$.

Proof:

Step 1: The rotation path. For $\theta \in [0, \pi/2]$ set

$$R_\theta = \begin{pmatrix} \cos \theta 1_n & -\sin \theta 1_n \\ \sin \theta 1_n & \cos \theta 1_n \end{pmatrix} \in \mathcal{U}_{2n}(A)$$

a unitary because $R_\theta R_\theta^* = 1_{2n}$ by the identity $\cos^2 \theta + \sin^2 \theta = 1$ (the scalar entries commute with everything). The map $\theta \mapsto R_\theta$ is norm-continuous, $R_0 = 1_{2n}$, and

$$R_{\pi/2} = \begin{pmatrix} 0 & -1_n \\ 1_n & 0 \end{pmatrix}$$

Now consider, for $u \in \mathcal{U}_n(A)$,

$$W(\theta) = \begin{pmatrix} u & 0 \\ 0 & 1_n \end{pmatrix} R_\theta \begin{pmatrix} u^* & 0 \\ 0 & 1_n \end{pmatrix} R_\theta^*$$

Each $W(\theta)$ is a product of unitaries, hence unitary, and $\theta \mapsto W(\theta)$ is continuous. At $\theta = 0$, $W(0) = 1_{2n}$. At $\theta = \pi/2$ direct computation gives

$$\begin{pmatrix} u & 0 \\ 0 & 1_n \end{pmatrix} \begin{pmatrix} 0 & -1_n \\ 1_n & 0 \end{pmatrix} \begin{pmatrix} u^* & 0 \\ 0 & 1_n \end{pmatrix} \begin{pmatrix} 0 & 1_n \\ -1_n & 0 \end{pmatrix} = \begin{pmatrix} u & 0 \\ 0 & u^* \end{pmatrix}$$

where $R_{\pi/2}^* = \begin{pmatrix} 0 & 1_n \\ -1_n & 0 \end{pmatrix}$ and carrying out the four block multiplications left to right yields $\text{diag}(u, u^*)$. Thus $\theta \mapsto W(\theta)$ is a path of unitaries from 1_{2n} to $\text{diag}(u, u^*)$, proving (1).

Step 2: From the product to the block sum. It suffices to show that both $\text{diag}(uv, 1)$ and $\text{diag}(u, v)$ are connected, in $\mathcal{U}_{2n}(A)$, to the common unitary $\text{diag}(1, uv)$. Two factorizations achieve this. First,

$$\begin{pmatrix} uv & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} uv & 0 \\ 0 & (uv)^* \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & uv \end{pmatrix}$$

since $(uv)^*(uv) = 1$; the first factor lies in $\mathcal{U}_{2n}(A)_0$ by (1) applied to uv , joined to 1_{2n} by a path $p(t)$ of unitaries, so $t \mapsto p(t)\text{diag}(1, uv)$ joins $\text{diag}(uv, 1)$ to $\text{diag}(1, uv)$. Second,

$$\begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix} = \begin{pmatrix} u & 0 \\ 0 & u^* \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & uv \end{pmatrix}$$

since $u^* \cdot uv = v$; the first factor lies in $\mathcal{U}_{2n}(A)_0$ by (1) applied to u , joined to 1_{2n} by a path $q(t)$, so $t \mapsto q(t)\text{diag}(1, uv)$ joins $\text{diag}(u, v)$ to $\text{diag}(1, uv)$. Composing one path with the reverse of the other connects $\text{diag}(uv, 1)$ to $\text{diag}(u, v)$ inside $\mathcal{U}_{2n}(A)$, completing the proof.

The Whitehead lemma is the analytic heart of K_1 . Part (1) shows that any unitary, when paired with its adjoint in a doubled matrix algebra, becomes contractible to the identity; this is the unitary analogue of the algebraic fact that $\text{diag}(g, g^{-1})$ is a product of elementary matrices. Part (2) shows that the block sum operation and the group multiplication agree after passage to path components. With it in hand, the group structure of $K_1(A)$ and its commutativity follow at once.

Theorem 1.2.4: $K_1(A)$ is an Abelian Group

Let A be a unital C^* -algebra. Then $K_1(A) = \pi_0(\mathcal{U}_\infty(A))$, with the operation of definition 1.2.1, is an abelian group. For unitaries u, v (placed in a common $\mathcal{U}_n(A)$ after stabilizing),

$$[u]_1 + [v]_1 = [uv]_1, \quad -[u]_1 = [u^*]_1, \quad [1]_1 = 0$$

and $[u]_1 = 0$ if and only if $u \oplus 1_m \in \mathcal{U}_{n+m}(A)_0$ for some $m \geq 0$.

Proof:

Step 1: Well-definedness of the operation. The block sum $[u]_1 + [v]_1 = [u \oplus v]_1$ does not depend on representatives: if $u' = u_0 u$ with $u_0 \in \mathcal{U}_n(A)_0$ joined to 1 by a path $p(t)$, then $u' \oplus v$ is joined to $u \oplus v$ by $t \mapsto (p(t) \oplus 1)(u \oplus v)$, so $[u' \oplus v]_1 = [u \oplus v]_1$, and symmetrically in v . It is also independent of the matrix size into which u and v are placed, since enlarging u to $u \oplus 1$ does not change its class (the inclusion $\mathcal{U}_n \hookrightarrow \mathcal{U}_{n+1}$ is built into $\mathcal{U}_\infty$), and similarly for v . Thus $+$ is a well-defined binary operation on $K_1(A)$.

Step 2: The operation equals the group product. For $u, v \in \mathcal{U}_n(A)$, lemma 1.2.3(2) gives a path in $\mathcal{U}_{2n}(A)$ from $u \oplus v$ to $(uv) \oplus 1$. Hence in $K_1(A)$,

$$[u]_1 + [v]_1 = [u \oplus v]_1 = [(uv) \oplus 1]_1 = [uv]_1$$

So the induced operation on path components coincides with the (componentwise) group multiplication of $\mathcal{U}_\infty(A)$.

Step 3: Group axioms. Associativity and the identity element are inherited from the group multiplication of $\mathcal{U}_\infty(A)$ via Step 2: $[1]_1$ is the identity because $1 \cdot u = u$, and $([u]_1 + [v]_1) + [w]_1 = [uvw]_1 = [u]_1 + ([v]_1 + [w]_1)$. For inverses, $[u]_1 + [u^*]_1 = [uu^*]_1 = [1]_1 = 0$, so $-[u]_1 = [u^*]_1$. Therefore $K_1(A)$ is a group, and $\mathcal{U}_\infty(A)_0$ is exactly the kernel of $\mathcal{U}_\infty(A) \rightarrow K_1(A)$, hence a normal subgroup, confirming the quotient description in definition 1.2.1.

Step 4: Commutativity. By Step 2 and lemma 1.2.3(2) applied with the roles of u, v exchanged,

$$[u]_1 + [v]_1 = [uv]_1 = [u \oplus v]_1 \quad \text{and} \quad [v]_1 + [u]_1 = [vu]_1 = [v \oplus u]_1$$

But $u \oplus v$ and $v \oplus u$ are conjugate in $\mathcal{U}_{2n}(A)$ by the swap unitary $\Sigma = R_{\pi/2}$ of lemma 1.2.3, and $\Sigma \in \mathcal{U}_{2n}(A)_0$ via the path $\theta \mapsto R_\theta$. Writing $\Sigma = p(1)$ for that path with $p(0) = 1$, the map $t \mapsto p(t)(u \oplus v)p(t)^*$ is a path of unitaries from $u \oplus v$ to $\Sigma(u \oplus v)\Sigma^* = v \oplus u$. Hence $[u \oplus v]_1 = [v \oplus u]_1$, giving $[u]_1 + [v]_1 = [v]_1 + [u]_1$.

Step 5: The triviality criterion. By definition $[u]_1 = 0$ means u lies in the identity component of $\mathcal{U}_\infty(A)$, which is the union of the $\mathcal{U}_{n+m}(A)_0$; a path in $\mathcal{U}_\infty(A)$ has compact image, hence lands in some $\mathcal{U}_{n+m}(A)$, so $[u]_1 = 0$ if and only if $u \oplus 1_m \in \mathcal{U}_{n+m}(A)_0$ for some m , as we sought to show.

The theorem completes the foundational picture: $K_1(A)$ is a genuine abelian group whose addition is realized either by the block sum (geometrically natural) or by matrix multiplication (algebraically natural), the two being identified by the Whitehead lemma. The commutativity is forced not by any commutativity in A but by the freedom to swap blocks in a large enough matrix algebra, exactly the Eckmann–Hilton phenomenon: two compatible unital binary operations on a set must agree and be commutative. Here the block sum and the matrix product are the two operations, and the swap path provides the interchange.

The first concrete computation grounds the definition: the simplest unital C^* -algebra has trivial K_1 .

Example 1.2.5: $K_1(\mathbb{C}) = 0$

Take $A = \mathbb{C}$. Then $M_n(\mathbb{C})$ is the algebra of $n \times n$ complex matrices, and $\mathcal{U}_n(\mathbb{C}) = U(n)$ is the classical unitary group. Every unitary matrix $u \in U(n)$ has the form $u = e^{ih}$ for a self-adjoint h : diagonalize $u = w \operatorname{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) w^*$ with $w \in U(n)$ and real θ_j (by the spectral theorem for normal matrices), and set $h = w \operatorname{diag}(\theta_1, \dots, \theta_n) w^*$, which is self-adjoint with $e^{ih} = u$. By lemma 1.2.2, the path $t \mapsto e^{it h}$ joins 1 to u inside $U(n)$. Hence $U(n)$ is path-connected for every n , so each $\mathcal{U}_n(\mathbb{C})$ has a single path component, and therefore

$$K_1(\mathbb{C}) = \pi_0(\mathcal{U}_\infty(\mathbb{C})) = \pi_0\left(\varinjlim_n U(n)\right) = 0$$

The connectivity also follows from $\det : U(n) \rightarrow \mathbb{T}$ being a surjective fibration with connected fiber $SU(n)$ and connected base $\mathbb{T}$. This matches the topological side: $\mathbb{C} = C(\text{pt})$, and $K^1(\text{pt}) = \tilde{K}^0(\Sigma \text{pt}) = \tilde{K}^0(\text{pt}) = 0$.

The computation $K_1(\mathbb{C}) = 0$ generalizes immediately to matrix algebras and, by a stabilization argument, places K_1 on the same footing as K_0 with respect to functoriality and Morita-type invariance. The next results record these structural properties before the central commutative example $K_1(C(S^1)) = \mathbb{Z}$.

Theorem 1.2.6: Functoriality of K_1

Let $\varphi : A \rightarrow B$ be a unital $*$ -homomorphism of unital C^* -algebras. Then φ induces a group homomorphism

$$K_1(\varphi) = \varphi_* : K_1(A) \rightarrow K_1(B), \quad \varphi_*[u]_1 = [\varphi_n(u)]_1$$

for $u \in \mathcal{U}_n(A)$, where $\varphi_n : M_n(A) \rightarrow M_n(B)$ is the entrywise application of φ . The assignment $A \mapsto K_1(A)$, $\varphi \mapsto \varphi_*$ is a covariant functor from unital C^* -algebras to abelian groups: $(\operatorname{id}_A)_* = \operatorname{id}_{K_1(A)}$ and $(\psi\varphi)_* = \psi_*\varphi_*$. Moreover K_1 is homotopy invariant: if $\varphi_0, \varphi_1 : A \rightarrow B$ are homotopic through a path $t \mapsto \varphi_t$ of unital $*$ -homomorphisms (continuous in the sense that $t \mapsto \varphi_t(a)$ is norm-continuous for each $a \in A$), then $(\varphi_0)_* = (\varphi_1)_*$.

Proof:

Step 1: φ_n preserves unitaries and is continuous. A unital $*$ -homomorphism satisfies $\varphi(a^*) = \varphi(a)^*$ and $\varphi(1_A) = 1_B$, so $\varphi_n(u)^* \varphi_n(u) = \varphi_n(u^* u) = \varphi_n(1_n) = 1_n$ and likewise $\varphi_n(u) \varphi_n(u)^* = 1_n$; thus φ_n maps $\mathcal{U}_n(A)$ into $\mathcal{U}_n(B)$. Every $*$ -homomorphism between C^* -algebras is norm-decreasing, hence continuous (this is a standard consequence of the C^* -identity, established in

[4]), so φ_n is continuous and the maps φ_n are compatible with the block-diagonal inclusions, assembling to a continuous group homomorphism $\varphi_\infty : \mathcal{U}_\infty(A) \rightarrow \mathcal{U}_\infty(B)$.

Step 2: Induced map on path components. A continuous group homomorphism sends path components to path components: if u, u' are joined by a path $p(t)$ in $\mathcal{U}_\infty(A)$, then $\varphi_\infty \circ p$ joins $\varphi_\infty(u)$ to $\varphi_\infty(u')$ in $\mathcal{U}_\infty(B)$. Hence $\varphi_*[u]_1 := [\varphi_\infty(u)]_1$ is well-defined, and it is a homomorphism because $\varphi_\infty(u \oplus v) = \varphi_\infty(u) \oplus \varphi_\infty(v)$. Functoriality $(\text{id})_* = \text{id}$ and $(\psi\varphi)_* = \psi_*\varphi_*$ is immediate from $(\text{id})_\infty = \text{id}$ and $(\psi\varphi)_\infty = \psi_\infty\varphi_\infty$.

Step 3: Homotopy invariance. Let $u \in \mathcal{U}_n(A)$ and let $t \mapsto \varphi_t$ be a homotopy. The map $t \mapsto (\varphi_t)_n(u) \in \mathcal{U}_n(B)$ is continuous: each entry $((\varphi_t)_n(u))_{ij} = \varphi_t(u_{ij})$ is norm-continuous in t by hypothesis, and continuity entrywise is continuity in $M_n(B)$ (recall the matrix-algebra norm controls and is controlled by the entry norms). Each $(\varphi_t)_n(u)$ is unitary by Step 1 applied to φ_t . Thus $t \mapsto (\varphi_t)_n(u)$ is a path of unitaries in $\mathcal{U}_n(B)$ from $(\varphi_0)_n(u)$ to $(\varphi_1)_n(u)$, so $[(\varphi_0)_n(u)]_1 = [(\varphi_1)_n(u)]_1$, that is $(\varphi_0)_*[u]_1 = (\varphi_1)_*[u]_1$, completing the proof.

Functoriality and homotopy invariance together make K_1 a homotopy-theoretic invariant of C^* -algebras, exactly parallel to K_0 . Homotopy invariance has an immediate consequence: a C^* -algebra that is contractible (homotopy equivalent to 0, or to $\mathbb{C}$ in the unital setting) has K_1 equal to that of the model. The next property, stability, says K_1 cannot see the difference between A and its matrix amplifications, which is what justifies having taken a colimit over all matrix sizes in the definition.

Theorem 1.2.7: Stability of K_1

Let A be a unital C^* -algebra and $m \geq 1$. The block-diagonal embedding $\lambda : A \rightarrow M_m(A)$, $a \mapsto a \oplus 0_{m-1} = \text{diag}(a, 0, \dots, 0)$, sending 1_A to a rank-one projection, induces an isomorphism

$$K_1(M_m(A)) \cong K_1(A)$$

More precisely, the inclusion $M_n(A) \hookrightarrow M_n(M_m(A)) = M_{nm}(A)$ identifies $\mathcal{U}_\infty(M_m(A))$ with a cofinal part of $\mathcal{U}_\infty(A)$, inducing the isomorphism on π_0 .

Proof:

There is a $*$ -isomorphism $M_n(M_m(A)) \cong M_{nm}(A)$ for every n , given by regarding an $n \times n$ matrix of $m \times m$ matrices over A as a single $nm \times nm$ matrix over A ; this is the standard “forgetting the block partition” identification, and it is a unital $*$ -isomorphism. These identifications are compatible with the block-diagonal inclusions defining the two infinite unitary groups: increasing n by one for $M_m(A)$ increases the underlying size by m for A . Hence the system $\{\mathcal{U}_{nm}(A) \mid n \geq 1\}$, a cofinal subsystem of $\{\mathcal{U}_k(A) \mid k \geq 1\}$ (every k is dominated by some nm), has the same colimit:

$$\mathcal{U}_\infty(M_m(A)) = \varinjlim_n \mathcal{U}_n(M_m(A)) = \varinjlim_n \mathcal{U}_{nm}(A) = \varinjlim_k \mathcal{U}_k(A) = \mathcal{U}_\infty(A)$$

where the middle equality uses that a cofinal subsystem of a directed system has the same colimit, and each map is a homeomorphism onto its image. Taking π_0 , which commutes with the colimit because path components of a colimit of inclusions are computed levelwise, yields $K_1(M_m(A)) \cong K_1(A)$. The embedding λ induces this isomorphism since on the level of $\mathcal{U}_n$ it is exactly the inclusion $\mathcal{U}_n(A) = \mathcal{U}_n(\lambda(A)) \hookrightarrow \mathcal{U}_n(M_m(A))$ followed by the identification above, as we sought to show.

Stability instantly furnishes a family of examples: every full matrix algebra over $\mathbb{C}$ has trivial K_1 .

Example 1.2.8: $K_1(M_n(\mathbb{C})) = 0$

Apply theorem 1.2.7 with $A = \mathbb{C}$ and $m = n$: $K_1(M_n(\mathbb{C})) \cong K_1(\mathbb{C}) = 0$ by example 1.2.5. Directly, $\mathcal{U}_k(M_n(\mathbb{C})) = U(kn)$ is path-connected for all k (the spectral-theorem argument of example 1.2.5), so $\pi_0(\mathcal{U}_\infty(M_n(\mathbb{C}))) = 0$. The same reasoning gives K_1 of any finite-dimensional C^* -algebra: such an algebra is a finite direct sum $\bigoplus_i M_{n_i}(\mathbb{C})$ by the Artin–Wedderburn classification of finite-dimensional C^* -algebras (see [4]), and K_1 of a direct sum is the direct sum of the K_1 's (the unitary group of a product is the product of unitary groups, and π_0 of a product of these is the product of π_0 's), so $K_1(\bigoplus_i M_{n_i}(\mathbb{C})) = \bigoplus_i K_1(M_{n_i}(\mathbb{C})) = 0$.

The two examples so far have $K_1 = 0$ because their unitary groups are connected. The phenomenon that makes K_1 nontrivial is a disconnected unitary group, and the prototype is the circle algebra $C(S^1)$, where the obstruction to connectivity is the winding number. This is the bridge to the topological group $K^1(S^1)$, and the computation is recorded in full.

Example 1.2.9: $K_1(C(S^1)) \cong \mathbb{Z}$ via Winding Number

Let $A = C(S^1)$, the C^* -algebra of continuous complex-valued functions on the circle $S^1 = \mathbb{T}$, with pointwise operations, complex conjugation as involution, and the supremum norm. A unitary $u \in \mathcal{U}_1(A) = \mathcal{U}(C(S^1))$ is a continuous function $u : S^1 \rightarrow \mathbb{C}$ with $u(z)\overline{u(z)} = 1$ for all z , that is a continuous map $u : S^1 \rightarrow \mathbb{T}$. The set of homotopy classes of such maps is computed in [5]: a continuous map $S^1 \rightarrow \mathbb{T} \simeq S^1$ has a well-defined *winding number* (degree)

$$w(u) = \frac{1}{2\pi i} \oint_{S^1} \frac{u'(z)}{u(z)} dz \in \mathbb{Z}$$

when u is smooth, and in general by its homotopy class, and two such maps are homotopic if and only if they have equal winding number, giving $\pi_0(\mathcal{U}_1(A)) \cong [S^1, \mathbb{T}] \cong \pi_1(S^1) \cong \mathbb{Z}$, generated by the identity function $z \mapsto z$. The claim is that stabilizing to $\mathcal{U}_\infty$ does not change this.

Stabilization does not enlarge the invariant. For $u \in \mathcal{U}_n(C(S^1))$, the determinant $\det : \mathcal{U}_n(C(S^1)) \rightarrow \mathcal{U}(C(S^1))$, $u \mapsto \det u$ (the pointwise determinant, a continuous $S^1 \rightarrow \mathbb{T}$ since $u(z) \in U(n)$ and $\det : U(n) \rightarrow \mathbb{T}$ is continuous), is a group homomorphism compatible with block sums: $\det(u \oplus v) = \det(u) \det(v)$, and $\det(u \oplus 1) = \det(u)$. Composing with the winding number gives a homomorphism

$$W : \mathcal{U}_\infty(C(S^1)) \rightarrow \mathbb{Z}, \quad W(u) = w(\det u)$$

which is continuous (the winding number is locally constant on the space of maps $S^1 \rightarrow \mathbb{T}$), hence factors through π_0 to give $\bar{W} : K_1(C(S^1)) \rightarrow \mathbb{Z}$. It is surjective because $z \mapsto z$ in $\mathcal{U}_1$ has $W = 1$. For injectivity, suppose $W(u) = 0$ for $u \in \mathcal{U}_n(C(S^1))$, viewed as a map $u : S^1 \rightarrow U(n)$. Then $\det u : S^1 \rightarrow \mathbb{T}$ has winding number 0, so it lifts along the covering $\mathbb{R} \rightarrow \mathbb{T}$, $\theta \mapsto e^{i\theta}$, to a continuous $\theta : S^1 \rightarrow \mathbb{R}$ with $\det u = e^{i\theta}$. Set $d = \text{diag}(\det u, 1, \dots, 1) \in \mathcal{U}_n(C(S^1))$; the map $t \mapsto \text{diag}(e^{it\theta}, 1, \dots, 1)$ joins d to 1, so $[d]_1 = 0$. The product $w = u d^{-1}$ takes values in $SU(n)$, since $\det w = \det u \cdot (\det u)^{-1} = 1$ pointwise. Because $SU(n)$ is simply connected, $\pi_1(SU(n)) = 0$, so the map $w : S^1 \rightarrow SU(n)$ is null-homotopic, hence connected to the constant map 1 through a path of $SU(n)$ -valued unitaries, giving $[w]_1 = 0$. Therefore $[u]_1 = [w]_1 + [d]_1 = 0$, so $\bar{W}$ is injective. Hence

$$K_1(C(S^1)) \cong \mathbb{Z}$$

generated by the class of the identity function $u(z) = z$, the winding number being the isomorphism.

The winding-number computation is the operator-theoretic shadow of a topological fact, and it is precisely the bridge to topological K -theory promised in the section opening. The identification is recorded as the central theorem of the section.

Theorem 1.2.10: K_1 of a Commutative C^* -Algebra

Let X be a compact Hausdorff space and $A = C(X)$ the C^* -algebra of continuous complex-valued functions on X . Then there is a natural isomorphism

$$K_1(C(X)) \cong K^1(X) := \tilde{K}^0(\Sigma X)$$

where ΣX is the (reduced) suspension of X and $K^1(X)$ is the topological K^1 of [2]. Under this isomorphism the class $[u]_1$ of a unitary $u \in \mathcal{U}_n(C(X))$ corresponds to the homotopy class of the map $u : X \rightarrow U(n) \subset U$, viewed as an element of $[X, U] = \tilde{K}^0(\Sigma X)$.

Proof:

Step 1: Unitaries are maps into the unitary group. A unitary $u \in \mathcal{U}_n(C(X))$ is exactly a continuous function $u : X \rightarrow M_n(\mathbb{C})$ with $u(x) \in U(n)$ for every x , that is a continuous map $u : X \rightarrow U(n)$. This is because the C^* -algebra $M_n(C(X)) \cong C(X, M_n(\mathbb{C}))$ is the algebra of continuous $M_n(\mathbb{C})$ -valued functions (matrix entries are continuous scalar functions), and the unitary condition $u^*u = uu^* = 1$ holds in $C(X, M_n(\mathbb{C}))$ if and only if it holds pointwise. Block sum of unitaries corresponds to the map $x \mapsto u(x) \oplus v(x)$ into $U(n+m)$, and the stabilization $\mathcal{U}_n \hookrightarrow \mathcal{U}_{n+1}$ corresponds to the standard inclusion $U(n) \hookrightarrow U(n+1)$.

Step 2: Path components are homotopy classes into U . Two unitaries $u, u' \in \mathcal{U}_n(C(X))$ lie in the same path component of $\mathcal{U}_n(C(X))$ if and only if they are homotopic as maps $X \rightarrow U(n)$: a path of unitaries $t \mapsto p(t) \in \mathcal{U}_n(C(X))$ is precisely a homotopy $(x, t) \mapsto p(t)(x) \in U(n)$, using that continuity of $t \mapsto p(t)$ in the norm of $C(X, M_n(\mathbb{C}))$ is uniform continuity in x , hence joint continuity of $(x, t) \mapsto p(t)(x)$, and conversely. Passing to the colimit over n ,

$$K_1(C(X)) = \pi_0 \left(\varinjlim_n \mathcal{U}_n(C(X)) \right) = \varinjlim_n [X, U(n)] = [X, U]$$

where $U = \varinjlim_n U(n)$ and the colimit of homotopy-class sets is taken along the stabilization inclusions; the second equality uses that a path in $\varinjlim_n \mathcal{U}_n(C(X))$ has image in some $\mathcal{U}_n(C(X))$ by compactness of $[0, 1]$, so π_0 commutes with the colimit.

Step 3: Free maps as based maps off a disjoint basepoint. Step 2 produced $K_1(C(X)) = [X, U]$ with *free* (not basepoint-preserving) homotopy classes, because a unitary $u \in \mathcal{U}_n(C(X))$ is an arbitrary continuous map $X \rightarrow U(n)$ with no constraint at any chosen point of X . To match it to the topological K^1 , which is built from *based* classes, adjoin a disjoint basepoint. A continuous map $X \rightarrow U$ is the same datum as a based continuous map $X_+ = X \sqcup \{*\} \rightarrow U$ sending the added point $*$ to the identity $1 \in U$: extending a free map by $* \mapsto 1$ gives a based map, and restricting a based map $X_+ \rightarrow U$ to X gives back a free map, and the two operations are mutually inverse and respect homotopies (a free homotopy of maps $X \rightarrow U$ extends to a based homotopy of maps $X_+ \rightarrow U$ by sending $*$ to 1 throughout). Hence

$$K_1(C(X)) = [X, U] = [X_+, U]_*$$

an identity of *based* homotopy-class sets that uses no connectedness or CW hypothesis on X .

Step 4: The based classes are $K^{-1}(X)$. By the representability and loop-space picture of [2], for any pointed compact space Y one has $\tilde{K}^0(\Sigma Y) \cong [\Sigma Y, BU \times \mathbb{Z}]_* \cong [Y, \Omega(BU \times \mathbb{Z})]_*$, the second isomorphism being the loop-suspension adjunction. Now $\Omega(BU \times \mathbb{Z}) \simeq \Omega(BU) \simeq U$, since $\mathbb{Z}$ is discrete (so $\Omega\mathbb{Z}$ is a point) and the loop space of the classifying space BU is the group U itself (the classifying-space property, established in [5]). Applying this with the pointed space $Y = X_+$ gives $[X_+, U]_* \cong \tilde{K}^0(\Sigma(X_+))$, which by [2] is exactly the unreduced negative group $K^{-1}(X) = \tilde{K}^0(\Sigma(X_+))$. Combining with Step 3,

$$K_1(C(X)) = [X_+, U]_* = \tilde{K}^0(\Sigma(X_+)) = K^{-1}(X) = K^1(X)$$

the last equality being the periodic-grading convention $K^{-1} = K^1$ of [2].

Step 5: Reconciling with $\tilde{K}^0(\Sigma X)$. It remains to identify $\tilde{K}^0(\Sigma(X_+))$ with the reduced form $\tilde{K}^0(\Sigma X)$ recorded in the theorem statement. The suspension of a space with a disjoint basepoint splits off a circle: there is a homotopy equivalence

$$\Sigma(X_+) \simeq \Sigma X \vee S^1$$

because $X_+ = X \sqcup \{*\}$ and suspending the disjoint union of X with a point yields the wedge of ΣX with $\Sigma S^0 = S^1$ (the suspension coordinate over the added point sweeps out a circle wedged on at the basepoint). Reduced K^0 is additive on wedges of compact pointed spaces, $\tilde{K}^0(A \vee B) \cong \tilde{K}^0(A) \oplus \tilde{K}^0(B)$ (the wedge axiom for the reduced theory, [2]), so

$$\tilde{K}^0(\Sigma(X_+)) \cong \tilde{K}^0(\Sigma X) \oplus \tilde{K}^0(S^1)$$

The second summand vanishes: S^1 is an odd sphere, and $\tilde{K}^0(S^1) = K^{-1}(\text{pt}) = 0$ by the sphere computation ([2]), since every complex vector bundle over S^1 is stably trivial via the clutching correspondence ([2]). Therefore $\tilde{K}^0(\Sigma(X_+)) \cong \tilde{K}^0(\Sigma X)$, and chaining with Step 4,

$$K_1(C(X)) = \tilde{K}^0(\Sigma(X_+)) = \tilde{K}^0(\Sigma X) = K^1(X)$$

exactly the topological $K^1(X)$ of theorem 1.2.10's statement, with no connectedness assumption imposed anywhere.

Step 6: Naturality. A continuous map $f : Y \rightarrow X$ induces the unital $*$ -homomorphism $f^* : C(X) \rightarrow C(Y)$, $g \mapsto g \circ f$, hence $K_1(f^*) : K_1(C(X)) \rightarrow K_1(C(Y))$; under the identification of Steps 1–5 this sends the class of $u : X \rightarrow U$ to the class of $u \circ f : Y \rightarrow U$, which is exactly the topological map $K^1(X) \rightarrow K^1(Y)$ induced by f . So the isomorphism is natural in X , completing the proof.

The theorem closes the circle between the two halves of this book. The operator-theoretic $K_1(C(X))$, built from path components of unitary groups of matrix algebras, is the topological $K^1(X)$, built from vector bundles over the suspension; the winding number of example 1.2.9 is the special case $X = S^1$, where $[S^1, U] = \pi_1(U) = \mathbb{Z}$ by [2]. The contravariant functor $X \mapsto C(X)$ of Gelfand–Naimark duality (theorem 1.0.3) thus intertwines topological and operator K^1 , just as it does for K^0 . With the commutative case settled, two further computations complete the catalogue of basic examples: the non-unital algebra of compact operators, and the reduction of K_1 for non-unital algebras in general.

For a non-unital C^* -algebra A there is no unitary group to speak of (there is no 1), so K_1 is defined by passing to the unitization $\tilde{A}$ of proposition 1.0.4 and removing the contribution of the scalars

$$\mathbb{C} = \tilde{A}/A.$$

Definition 1.2.11: K_1 for Non-Unital C^* -Algebras

Let A be a (not necessarily unital) C^* -algebra with unitization $\tilde{A}$ and quotient map $\pi : \tilde{A} \rightarrow \tilde{A}/A \cong \mathbb{C}$. The K_1 -group of A is

$$K_1(A) = \ker(\pi_* : K_1(\tilde{A}) \rightarrow K_1(\mathbb{C})) = K_1(\tilde{A})$$

where the second equality holds because $K_1(\mathbb{C}) = 0$ by example 1.2.5, so the kernel is all of $K_1(\tilde{A})$. For unital A this agrees with definition 1.2.1 up to the natural splitting $\tilde{A} \cong A \oplus \mathbb{C}$.

Because $K_1(\mathbb{C}) = 0$, the non-unital definition reduces to $K_1(A) = K_1(\tilde{A})$, which is one reason K_1 is technically smoother than K_0 (where the analogous kernel is genuinely a proper subgroup and the scalar correction is essential, as the non-exactness noted after definition 1.1.1 shows). The last example computes K_1 of the compact operators, the non-unital algebra most central to operator K -theory.

Example 1.2.12: $K_1(\mathcal{K}) = 0$ for the Compact Operators

Let $\mathcal{K} = \mathcal{K}(H)$ be the C^* -algebra of compact operators on a separable infinite-dimensional Hilbert space H . It is non-unital, so $K_1(\mathcal{K}) = K_1(\tilde{\mathcal{K}})$ by definition 1.2.11. The computation rests on continuity of K -theory under inductive limits.

Step 1: $\mathcal{K}$ as an inductive limit. The finite-rank operators are dense in $\mathcal{K}$, and choosing an orthonormal basis of H identifies the rank- $\leq n$ part with $M_n(\mathbb{C})$ via the top-left corner inclusions $M_n(\mathbb{C}) \hookrightarrow M_{n+1}(\mathbb{C})$. Thus $\mathcal{K}$ is the inductive limit (norm closure of the increasing union) of the C^* -algebras $M_n(\mathbb{C})$,

$$\mathcal{K} = \overline{\bigcup_n M_n(\mathbb{C})} = \varinjlim_n M_n(\mathbb{C})$$

in the category of C^* -algebras.

Step 2: K_1 commutes with the limit. For any inductive limit $A = \varinjlim_n A_n$ of C^* -algebras, $K_1(A) = \varinjlim_n K_1(A_n)$. The reason is that a unitary $u \in \mathcal{U}_k(\tilde{A})$ and a path of unitaries connecting two such can each be approximated, to within the gap that keeps unitaries unitary under lemma 1.2.2, by elements with entries in some finite stage $\tilde{A}_N$; so every K_1 -class and every relation among classes is already visible at a finite stage. (This continuity is the operator- K -theory analogue of the algebraic limit lemma of [3]; a complete treatment is in the operator-algebra references.) Applying it with $A_n = M_n(\mathbb{C})$, and using $K_1(M_n(\mathbb{C})) = 0$ from example 1.2.8,

$$K_1(\mathcal{K}) = \varinjlim_n K_1(M_n(\mathbb{C})) = \varinjlim_n 0 = 0$$

matching the topological computation $K^1(\text{pt}) = 0$ under the slogan that $\mathcal{K}$ is “ K -theoretically a point.”

The catalogue $K_1(\mathbb{C}) = K_1(M_n(\mathbb{C})) = K_1(\mathcal{K}) = 0$ and $K_1(C(S^1)) = \mathbb{Z}$ exhibits the same pattern as topological K -theory: the invariant is trivial on the building blocks (points, matrices, compacts) and detects the circle through its winding. What remains is to make precise the relationship between this operator-theoretic K_1 and the algebraic Whitehead group of [3], since the two are defined by

superficially similar quotients of general linear groups but measure genuinely different things.

The purely algebraic K_1 of a ring R , defined in [3], is the abelianization of the stable general linear group,

$$K_1^{\text{alg}}(R) = \text{GL}(R)/E(R) = \text{GL}(R)/[\text{GL}(R), \text{GL}(R)]$$

where $\text{GL}(R) = \varinjlim_n \text{GL}_n(R)$ and $E(R)$ is the subgroup generated by elementary matrices, equal to the commutator subgroup by Whitehead's lemma (the algebraic counterpart of lemma 1.2.3). The operator-theoretic K_1 of a unital C^* -algebra A is instead

$$K_1(A) = \text{GL}_\infty(A)/\text{GL}_\infty(A)_0 = \mathcal{U}_\infty(A)/\mathcal{U}_\infty(A)_0$$

the quotient by the *connected component* of the identity, where the second equality uses the polar-decomposition retraction of invertibles onto unitaries (every invertible $a = u|a|$ with $|a| = (a^*a)^{1/2}$ positive invertible, and $t \mapsto u|a|^t$ connects a to its unitary part u). The two quotients differ in what they collapse: K_1^{alg} collapses the algebraically distinguished subgroup $E(R)$, while $K_1(A)$ collapses the topologically distinguished subgroup $\text{GL}_\infty(A)_0$.

1.2.13 Remark: Algebraic versus Operator K_1 For a unital C^* -algebra A there is always a natural surjection

$$K_1^{\text{alg}}(A) = \text{GL}_\infty(A)/E(A) \twoheadrightarrow \text{GL}_\infty(A)/\text{GL}_\infty(A)_0 = K_1(A)$$

because $E(A) \subseteq \text{GL}_\infty(A)_0$: every elementary matrix $1 + aE_{ij}$ ($i \neq j$) is connected to 1 through the path $t \mapsto 1 + taE_{ij}$ of invertibles, so the elementary subgroup lies in the identity component. The surjection is in general *not* injective. The topological component $\text{GL}_\infty(A)_0$ can be strictly larger than the algebraic commutator subgroup $E(A)$, because connectivity uses limits (paths, hence the norm topology) that the purely algebraic construction cannot see. Thus the operator-theoretic K_1 is a quotient of, and generally distinct from, the algebraic Whitehead group: it is the *topological* K_1 , sensitive to the analytic structure of A .

The distinction is sharpest in two regimes. For a field $\mathbb{C}$ (viewed as a C^* -algebra) the algebraic $K_1^{\text{alg}}(\mathbb{C}) = \mathbb{C}^\times$ by [3], since $\text{GL}(\mathbb{C})/E(\mathbb{C}) \cong \mathbb{C}^\times$ via the determinant; but the operator $K_1(\mathbb{C}) = 0$ by example 1.2.5, because $\mathbb{C}^\times = \text{GL}_1(\mathbb{C})$ is connected, so the entire group $\mathbb{C}^\times$ collapses into the identity component. Here the surjection $\mathbb{C}^\times = K_1^{\text{alg}}(\mathbb{C}) \twoheadrightarrow K_1(\mathbb{C}) = 0$ has enormous kernel, the whole connected group $\mathbb{C}^\times$. This is the precise sense in which the two theories diverge: the algebraic theory remembers the determinant in $\mathbb{C}^\times$, the operator theory forgets everything that can be continuously deformed away, and over $\mathbb{C}$ the multiplicative group is entirely deformable. The operator-theoretic K_1 is the right invariant for topology and index theory precisely because it discards this analytically inert information and retains only the discrete homotopical obstruction, which for $C(X)$ is the topological $K^1(X)$ of theorem 1.2.10.

Exercise 1.2.1

1. Let A be a unital C^* -algebra and $u \in \mathcal{U}_n(A)$ a unitary with $\|u - 1\| < 2$. Show directly, without invoking lemma 1.2.2, that $[u]_1 = 0$ in $K_1(A)$ by exhibiting an explicit path from u to 1. (Hint: the segment $t \mapsto (1 - t) + tu$ stays invertible; relate it to the unitary part.)
2. Prove that for unital C^* -algebras A, B ,

$$K_1(A \oplus B) \cong K_1(A) \oplus K_1(B)$$

(Hint: identify $\mathcal{U}_n(A \oplus B)$ with $\mathcal{U}_n(A) \times \mathcal{U}_n(B)$ and take π_0 .)

3. Compute $K_1(C(\text{pt}))$ and $K_1(C([0, 1]))$, and explain using homotopy invariance (theorem 1.2.6) why they are equal. What is $K_1(C(D^2))$ for the closed disk D^2 ?
4. Using theorem 1.2.10 and the suspension description $K^1(X) = \tilde{K}^0(\Sigma X)$, compute $K_1(C(S^2))$. (Hint: $[S^2, U] = \pi_2(U)$; use [2].)
5. Let $u, v \in \mathcal{U}_n(A)$. Show that the commutator uvu^*v^* satisfies $[uvu^*v^*]_1 = 0$ in $K_1(A)$, directly from theorem 1.2.4, and interpret this as the statement that $K_1(A)$ is the abelianized, stabilized unitary group.
6. Explain precisely why $K_1^{\text{alg}}(\mathbb{C}) = \mathbb{C}^\times$ but $K_1(\mathbb{C}) = 0$, identifying the kernel of the natural surjection $K_1^{\text{alg}}(\mathbb{C}) \rightarrow K_1(\mathbb{C})$ and the role of the norm topology.

1.3 The Six-Term Exact Sequence and Bott Periodicity

The two functors K_0 and K_1 built in the previous sections are not independent invariants. They are the two halves of a single periodic theory, glued together by the connecting maps of a short exact sequence of C^* -algebras. This section makes that gluing precise. We first introduce the suspension $SA = C_0(\mathbb{R}) \otimes A$, the C^* -algebraic analogue of the topological reduced suspension, and prove the dimension-shift isomorphism $K_1(A) \cong K_0(SA)$, which forces us to read K_1 as “ K_{-1} ”. We then take a short exact sequence $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ and construct two connecting homomorphisms: the index map $\partial_1: K_1(A/I) \rightarrow K_0(I)$, built by lifting a unitary to a partial isometry, and the exponential map $\partial_0: K_0(A/I) \rightarrow K_1(I)$, built by lifting a projection to a self-adjoint element and exponentiating. Assembling the half-exact sequences for K_0 and K_1 with these connecting maps yields the six-term cyclic exact sequence, the computational engine of the theory. Finally Bott periodicity, $K_0(A) \cong K_0(S^2A)$, closes the cycle into a genuinely 2-periodic theory matching topological K -theory, and the Toeplitz extension $0 \rightarrow \mathcal{K} \rightarrow \mathcal{T} \rightarrow C(S^1) \rightarrow 0$ shows the whole apparatus computing the Fredholm index of a Toeplitz operator as a winding number.

Throughout, A is a C^* -algebra (definition 1.0.1), not assumed unital, $\tilde{A}$ is its unitization (proposition 1.0.4), and K_0 is as in definition 1.1.1, extended to the non-unital case by $K_0(A) = \ker(K_0(\tilde{A}) \rightarrow K_0(\mathbb{C}))$. We write $\mathcal{U}_n(B)$ for the unitary group of $M_n(B)$ and $\mathcal{U}(B) = \bigcup_n \mathcal{U}_n(B)$ under the embeddings $u \mapsto \text{diag}(u, 1)$; the group $K_1(B)$ is $\mathcal{U}(\tilde{B})$ modulo the connected component of the identity (definition 1.2.1).

The path from K_0 to K_1 in topology is the suspension: $\tilde{K}(SX) \cong \tilde{K}^{-1}(X)$. To import this into the operator setting one needs the algebra whose “space” is the suspension. For a locally compact Hausdorff space X the suspension SX deletes two cone points, and continuous functions vanishing at infinity on $\mathbb{R} \times X$ are exactly $C_0(\mathbb{R}) \otimes C_0(X)$. This is the model to abstract.

Definition 1.3.1: Suspension of a C^* -Algebra

Let A be a C^* -algebra. The *suspension* of A is the C^* -algebra

$$SA = C_0(\mathbb{R}) \otimes A \cong C_0(\mathbb{R}, A)$$

the algebra of continuous A -valued functions on $\mathbb{R}$ vanishing at infinity, with pointwise operations and the supremum norm. Equivalently, $SA \cong \{f \in C([0, 1], A) : f(0) = f(1) = 0\}$, the A -valued loops based at 0. The *cone* of A is $CA = C_0((0, 1], A) = \{f \in C([0, 1], A) : f(0) = 0\}$.

The cone is the contractible building block: the homotopy $h_t(s) = f(ts)$ from the identity to the zero map shows CA is homotopy equivalent to 0. The suspension is the quotient $CA \rightarrow SA$ that records the missing endpoint, fitting into the *cone exact sequence*

$$0 \longrightarrow SA \longrightarrow CA \xrightarrow{\text{ev}_1} A \longrightarrow 0$$

where ev_1 evaluates a function of $C_0((0, 1], A)$ at the right endpoint $s = 1$. Suspension is a functor: a $*$ -homomorphism $\varphi: A \rightarrow B$ induces $S\varphi = \text{id}_{C_0(\mathbb{R})} \otimes \varphi$, and homotopic $*$ -homomorphisms induce homotopic ones, so S descends to K -theory.

Theorem 1.3.2: The Dimension-Shift Isomorphism

For every C^* -algebra A there is a natural isomorphism

$$\theta_A: K_1(A) \xrightarrow{\sim} K_0(SA)$$

In particular $K_1(A) \cong K_0(SA)$, so K_1 is the once-suspended K_0 , justifying the reading $K_1 = K_{-1}$.

Proof:

The proof identifies both groups with homotopy classes of loops of invertibles and uses the clutching-type correspondence between such loops and projections over the suspension. We may assume A is unital; the non-unital case follows by applying the unital result to $\hat{A}$ and restricting to the kernels of the augmentations, since S is exact and commutes with unitization up to the split $S\hat{A}/SA \cong SC$.

Step 1: $K_1(A)$ as based homotopy classes of unitary loops. By definition $K_1(A) = \varinjlim_n \mathcal{U}_n(A)/\mathcal{U}_n(A)_0$, where the subscript 0 denotes the identity component. Two unitaries $u, v \in \mathcal{U}_n(A)$ are equal in $K_1(A)$ if and only if $u \oplus 1_m$ and $v \oplus 1_m$ lie in the same path component of $\mathcal{U}_{n+m}(A)$ for some m . A continuous path in $\mathcal{U}_{n+m}(A)$ from w to 1 is precisely a based loop of unitaries when we close it up; concretely, given $u \in \mathcal{U}_n(A)$, choose any continuous path $w: [0, 1] \rightarrow \mathcal{U}_{2n}(A)$ with $w(0) = \text{diag}(u^*, u)$ and $w(1) = 1_{2n}$ (such a path exists because $\text{diag}(u^*, u)$ is connected to 1 inside $\mathcal{U}_{2n}(A)$ via the rotation

$$R_t = \begin{pmatrix} \cos(\frac{\pi t}{2}) & -\sin(\frac{\pi t}{2}) \\ \sin(\frac{\pi t}{2}) & \cos(\frac{\pi t}{2}) \end{pmatrix} \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos(\frac{\pi t}{2}) & \sin(\frac{\pi t}{2}) \\ -\sin(\frac{\pi t}{2}) & \cos(\frac{\pi t}{2}) \end{pmatrix} \begin{pmatrix} u^* & 0 \\ 0 & 1 \end{pmatrix}$$

which is a unitary for each t and runs from $R_0 = \text{diag}(uu^*, 1) = 1_{2n}$ at $t = 0$ to $R_1 = \text{diag}(u^*, u)$ at $t = 1$; reverse the parameter via $t \mapsto 1 - t$ to obtain a path from $\text{diag}(u^*, u)$ to 1_{2n} .

The companion element $\text{diag}(u, u^*)$ is then equally connected to 1_{2n} , being the swap-conjugate $\sigma \text{diag}(u^*, u) \sigma^*$ by the permutation unitary σ that exchanges the two blocks; σ lies in the identity component of $\mathcal{U}_{2n}(A)$ because the unitary group is path-connected to its block transpositions, an explicit path being $\sigma_t = \frac{1}{2} \begin{pmatrix} 1 + e^{i\pi(1-t)} & 1 - e^{i\pi(1-t)} \\ 1 - e^{i\pi(1-t)} & 1 + e^{i\pi(1-t)} \end{pmatrix}$ on each transposed pair, with $\sigma_0 = \sigma$ and $\sigma_1 = 1$).

Step 2: From unitary paths to projection loops over the suspension. Realize $SA = \{f \in C([0, 1], A) : f(0) = f(1) = 0\}$, so $\widetilde{SA} = \{f \in C([0, 1], \tilde{A}) : f(0) = f(1) \in \mathbb{C} \cdot 1\}$, functions whose two endpoints agree and are scalar. Write $e = \text{diag}(1_n, 0) \in M_{2n}(A)$. Given a unitary path $w : [0, 1] \rightarrow \mathcal{U}_{2n}(A)$ with $w(0) = 1_{2n}$ and with $w(1)$ commuting with e , the function

$$p_w(t) = w(t) e w(t)^* = w(t) \begin{pmatrix} 1_n & 0 \\ 0 & 0 \end{pmatrix} w(t)^*$$

is a projection in $M_{2n}(\widetilde{SA})$: it has the constant scalar value e at $t = 0$, and at $t = 1$ it equals $w(1) e w(1)^* = e$ because $w(1)$ commutes with e , so the two endpoints agree and are scalar, and p_w defines a class $[p_w] - [e] \in K_0(SA)$. The canonical such path attached to a unitary $u \in \mathcal{U}_n(A)$ is the rotation path of Step 1, built with the roles of u and u^* arranged so that it runs from $w_u(0) = 1_{2n}$ to $w_u(1) = \text{diag}(u, u^*)$ (the companion element to $\text{diag}(u^*, u)$, equally connected to 1_{2n} as noted in Step 1); its endpoint $\text{diag}(u, u^*)$ commutes with e . Set

$$\theta_A([u]) = [p_{w_u}] - [e] \in K_0(SA)$$

A homotopy of unitaries u gives a homotopy of the paths w_u and hence of the projections p_{w_u} , so this descends to a homomorphism $\theta_A : K_1(A) \rightarrow K_0(SA)$; additivity follows from $p_{w_u \oplus w_{u'}} = p_{w_u} \oplus p_{w_{u'}}$ after a permutation, and the rotation of Step 1 shows the class is independent of the chosen path realizing the contraction.

Step 3: Bijectivity via an explicit inverse. It remains to show θ_A is a bijection. The argument is direct: construct an inverse $\eta_A : K_0(SA) \rightarrow K_1(A)$ by framing a projection loop with a continuous family of unitaries and reading off the residual unitary at the closing endpoint, then verify both composites are the identity. The only analytic input is the local unitary equivalence $z = qp + (1 - q)(1 - p)$ of lemma 1.1.3, Step 1, propagated along the interval.

Step 3a: Framing a projection loop. A class in $K_0(SA)$ is represented by $[p] - [e_0]$ where $p \in M_m(\widetilde{SA})$ is a projection and e_0 is its constant scalar endpoint value; concretely p is a norm-continuous path $t \mapsto p(t)$ of projections in $M_m(\tilde{A})$ with $p(0) = p(1) = e_0$ a scalar projection of some rank k . After conjugating the whole loop by a fixed scalar unitary (which changes neither the class $[p] - [e_0]$ nor the construction below) we may take $e_0 = \text{diag}(1_k, 0_{m-k})$ in standard form. The claim is that there is a continuous path $z : [0, 1] \rightarrow \mathcal{U}_m(\tilde{A})$ with $z(0) = 1_m$ and $z(t) e_0 z(t)^* = p(t)$ for all t . Partition $[0, 1]$ as $0 = t_0 < t_1 < \dots < t_N = 1$ finely enough that $\|p(s) - p(t)\| < 1$ whenever s, t lie in a common subinterval $[t_{j-1}, t_j]$ (possible by uniform continuity of p on the compact interval). On $[t_{j-1}, t_j]$ set

$$g_j(t) = p(t) p(t_{j-1}) + (1 - p(t))(1 - p(t_{j-1}))$$

By the computation of lemma 1.1.3, $g_j(t) - 1 = (p(t) - p(t_{j-1}))(2p(t_{j-1}) - 1)$ has norm < 1 , so $g_j(t)$ is invertible with norm-continuous polar part $\zeta_j(t) = g_j(t) |g_j(t)|^{-1} \in \mathcal{U}_m(\tilde{A})$ satisfying $\zeta_j(t) p(t_{j-1}) \zeta_j(t)^* = p(t)$ and $\zeta_j(t_{j-1}) = 1_m$. Splice these by accumulating the completed earlier framings on the right: for $t \in [t_{j-1}, t_j]$ define

$$z(t) = \zeta_j(t) \zeta_{j-1}(t_{j-1}) \cdots \zeta_2(t_2) \zeta_1(t_1)$$

Each $z(t)$ is unitary, $t \mapsto z(t)$ is continuous (the values match at each breakpoint $t = t_{j-1}$ because $\zeta_j(t_{j-1}) = 1_m$), and $z(0) = \zeta_1(0) = 1_m$. Telescoping the conjugations from the right ($\zeta_1(t_1)$ carries $e_0 = p(t_0)$ to $p(t_1)$, then $\zeta_2(t_2)$ carries $p(t_1)$ to $p(t_2)$, and so on up to $\zeta_j(t)$ carrying $p(t_{j-1})$ to $p(t)$) gives $z(t) e_0 z(t)^* = p(t)$ throughout. This proves the framing claim.

Step 3b: The inverse map. With z as in Step 3a, evaluate at the closing endpoint: $z(1) e_0 z(1)^* = p(1) = e_0$, so $z(1)$ commutes with $e_0 = \text{diag}(1_k, 0_{m-k})$ and is therefore block-diagonal, $z(1) = \text{diag}(a, d)$ with $a \in \mathcal{U}_k(\tilde{A})$ and $d \in \mathcal{U}_{m-k}(\tilde{A})$. Define

$$\eta_A([p] - [e_0]) = [a]_1 \in K_1(\tilde{A}) = K_1(A)$$

using $K_1(\tilde{A}) \cong K_1(A)$ for unital A (theorem 1.2.4, since $K_1(\mathbb{C}) = 0$). This is well-defined. Any two framings z, z' of the same loop p differ by $r(t) = z'(t)^* z(t)$, a path in $\mathcal{U}_m(\tilde{A})$ with $r(0) = 1_m$ and $r(t) e_0 r(t)^* = e_0$ for all t (both framings conjugate e_0 to $p(t)$), so $r(t) = \text{diag}(a_r(t), d_r(t))$ is block-diagonal throughout and $t \mapsto a_r(t)$ is a path in $\mathcal{U}_k(\tilde{A})$ from 1_k to the upper block $a_r(1) = (a')^* a$ of $r(1) = z'(1)^* z(1)$; hence $a = a' a_r(1)$ with $a_r(1) \in \mathcal{U}_k(\tilde{A})_0$, giving $[a]_1 = [a']_1$. A homotopy of projection loops p_s frames to a homotopy of unitaries $z_s(1)$ by running Step 3a with a partition refined uniformly over the compact square (s, t) , so $[a]_1$ is unchanged under homotopy; and η_A is additive because the block-diagonal sum of loops frames by the block-diagonal sum of the z 's, giving upper block $a \oplus a'$.

Step 3c: $\eta_A \circ \theta_A = \text{id}$. Start from $[u] \in K_1(A)$ with $u \in \mathcal{U}_n(A)$, so by Step 2 the projection loop is $p_{w_u}(t) = w_u(t) e w_u(t)^*$ with $e = e_0 = \text{diag}(1_n, 0)$, $k = n$, $m = 2n$, $w_u(0) = 1_{2n}$, $w_u(1) = \text{diag}(u, u^*)$. The path $z(t) = w_u(t)$ is itself a framing of p_{w_u} : it satisfies $z(0) = 1_{2n}$ and $z(t) e_0 z(t)^* = w_u(t) e w_u(t)^* = p_{w_u}(t)$. Its closing value is $z(1) = w_u(1) = \text{diag}(u, u^*)$, block-diagonal with respect to e_0 with upper-left $n \times n$ block u . By Step 3b the class η_A produces is independent of the framing, so $\eta_A(\theta_A[u]) = [u]_1 = [u]$, and therefore $\eta_A \circ \theta_A = \text{id}_{K_1(A)}$.

Step 3d: Surjectivity of θ_A . Step 3c gives $\eta_A \circ \theta_A = \text{id}_{K_1(A)}$, so θ_A is injective and the retraction η_A is surjective; it remains to see θ_A is surjective, equivalently that η_A is injective. The framing already does most of this. Given $[p] - [e_0]$ with framing z and $z(1) = \text{diag}(a, d)$, the path z exhibits $z(1) \in \mathcal{U}_m(\tilde{A})_0$ (it is joined to 1_m by z itself), so $[a]_1 + [d]_1 = [z(1)]_1 = 0$ and hence $[d]_1 = -[a]_1$. Thus the lower block contributes nothing new, and the candidate preimage of $[p] - [e_0]$ under θ_A is $[a]_1$. The verification that $\theta_A([a]_1) = [p] - [e_0]$ amounts to constructing a homotopy, through projection loops over SA , from p (stabilized) to the standard Bott-type loop p_{w_a} of Step 2: one slides the framing z to the Whitehead framing w_a keeping the closing block a fixed. The construction is the operator-algebraic transcription of [2], that the based-homotopy class of a clutching function determines the bundle over a suspension and conversely ([2]); it is a single direct but lengthy homotopy bookkeeping with no conceptual content beyond Steps 3a–3c, and the full details are carried out in [6, Chapters 7–8] and [1, Chapter 9]. Granting it, θ_A is surjective.

Steps 3c and 3d show θ_A is injective with surjective image, so θ_A is a bijection, and being a homomorphism (Step 2) it is an isomorphism, with inverse η_A . Injectivity and the explicit inverse η_A rest only on the framing construction of Steps 3a–3c, with no appeal to the six-term exact sequence developed later; this is what makes the later derivations that invoke theorem 1.3.2 non-circular. For commutative $A = C(X)$ the whole statement is the topological clutching correspondence of [2] read through Gelfand duality, and the framing above is the operator-algebraic form of trivializing the clutched bundle over each cone of the suspension.

Step 4: Naturality. For a $*$ -homomorphism $\varphi: A \rightarrow B$, the diagram comparing θ_A and θ_B commutes because $S\varphi$ acts on p_w by φ entrywise and $K_1(\varphi)$ acts on $[w]$ by φ entrywise, and

both are computed from the same loop w . Thus θ is a natural isomorphism, completing the proof.

The dimension-shift isomorphism is the structural reason the operator theory is periodic rather than an unbounded tower. It says K_1 contains no information beyond K_0 of a derived algebra: every odd invariant is an even invariant of the suspension. Combined with Bott periodicity below, which identifies K_0 of the double suspension with K_0 itself, the two groups K_0 and K_1 exhaust the theory. The same isomorphism defines the negative-index groups uniformly by iterating the suspension, and shows they collapse modulo 2.

Definition 1.3.3: Higher and Negative K -Groups

For $n \geq 0$ define $K_n(A) = K_0(S^n A)$, where $S^0 A = A$ and $S^n A = S(S^{n-1} A)$. By theorem 1.3.2 this agrees for $n = 1$ with the K_1 of definition 1.2.1, and Bott periodicity (theorem 1.3.10) gives $K_n(A) \cong K_{n+2}(A)$, so the only groups are K_0 and K_1 .

Example 1.3.4: The Suspension of $\mathbb{C}$ and $K_*(\mathbb{C})$

Take $A = \mathbb{C}$. Then $S\mathbb{C} = C_0(\mathbb{R})$, the continuous functions on the line vanishing at infinity. Via the one-point compactification $\mathbb{R}^+ = S^1$ (so $\mathbb{R} \cong S^1 \setminus \{1\}$), $C_0(\mathbb{R}) \cong \{f \in C(S^1) : f(1) = 0\}$, the kernel of evaluation at the basepoint, and $K_0(C_0(\mathbb{R})) = \tilde{K}^0(\mathbb{R}^+) = \tilde{K}^0(S^1)$. The dimension-shift isomorphism gives

$$K_1(\mathbb{C}) \cong K_0(S\mathbb{C}) = K_0(C_0(\mathbb{R})) = \tilde{K}^0(S^1)$$

Now $\mathcal{U}_n(\mathbb{C}) = \mathrm{U}(n)$ is connected for every n , so $K_1(\mathbb{C}) = 0$, and indeed $\tilde{K}^0(S^1) = 0$ because every complex vector bundle over S^1 is trivial (the clutching function $S^0 \rightarrow \mathrm{U}(n)$ lands in a connected group, [2]). On the other side $K_0(\mathbb{C}) = \mathbb{Z}$, generated by [1]. Iterating, $K_0(C_0(\mathbb{R}^2)) = K_0(S^2\mathbb{C}) = \tilde{K}^0(S^2) = \mathbb{Z}$, generated by the Bott class, while $K_1(C_0(\mathbb{R}^2)) = 0$. Thus $K_*(\mathbb{C})$ is $\mathbb{Z}$ in even degree and 0 in odd degree, matching $K_*(\mathrm{pt})$ exactly.

A short exact sequence of C^* -algebras

$$0 \longrightarrow I \xrightarrow{\iota} A \xrightarrow{\pi} A/I \longrightarrow 0$$

(with I a closed two-sided ideal, ι the inclusion, π the quotient map) does not generally split, and K_0, K_1 individually are only half-exact: applying either functor produces a three-term exact sequence in the middle but no map back. The connecting maps repair this. The index map descends from K_1 of the quotient to K_0 of the ideal; its construction is the operator-algebraic incarnation of the Fredholm index, as the Toeplitz example will show.

Definition 1.3.5: The Index Map

Let $0 \rightarrow I \rightarrow A \xrightarrow{\pi} A/I \rightarrow 0$ be a short exact sequence of C^* -algebras and $u \in \mathcal{U}_n(\widetilde{A/I})$ a unitary representing a class $[u] \in K_1(A/I)$. The *index map*

$$\partial_1: K_1(A/I) \longrightarrow K_0(I)$$

is defined as follows. Lift the unitary $\text{diag}(u^*, u) \in \mathcal{U}_{2n}(\widetilde{A/I})$ to a unitary $w \in \mathcal{U}_{2n}(\tilde{A})$ with $\tilde{\pi}(w) = \text{diag}(u^*, u)$ (possible because $\text{diag}(u^*, u)$ lies in the identity component, hence lifts to a unitary by surjectivity of $\tilde{\pi}$ on identity components). Then

$$\partial_1[u] = \left[w \begin{pmatrix} 1_n & 0 \\ 0 & 0 \end{pmatrix} w^* \right] - \left[\begin{pmatrix} 1_n & 0 \\ 0 & 0 \end{pmatrix} \right] \in K_0(I)$$

where the difference of projections lies in $K_0(I)$ because $\tilde{\pi}$ sends it to $\text{diag}(u^*, u)\text{diag}(1_n, 0)\text{diag}(u, u^*) - \text{diag}(1_n, 0) = \text{diag}(1_n, 0) - \text{diag}(1_n, 0) = 0$, so the projection difference has entries in I . When the unitary u admits a lift to a partial isometry $v \in M_n(A)$ (an element with v^*v and vv^* projections and $\pi(v) = u$), this formula collapses to

$$\partial_1[u] = [1 - vv^*] - [1 - v^*v] \in K_0(I)$$

the difference of the two defect projections, both of which lie in $M_n(I)$.

The mechanism is the Halmos picture of a partial isometry. A unitary u in the quotient lifts only to a contraction $a \in A$ with $\pi(a) = u$ and $a^*a, aa^* \equiv 1$ modulo I ; the defect operators $1 - a^*a$ and $1 - aa^*$ measure the failure to be unitary and live in I . Doubling to $\text{diag}(u^*, u)$ and lifting to an honest unitary w converts these defects into a projection difference, and that difference is the $K_0(I)$ -valued index. When $A = \mathcal{T}$ is the Toeplitz algebra and $A/I = C(S^1)$, the lift is a Fredholm Toeplitz operator T_u and the projection difference $[1 - T_u T_u^*] - [1 - T_u^* T_u]$ is $[\ker T_u^*] - [\ker T_u]$, the negative of the analytic Fredholm index. The exponential map is the even-to-odd companion, built from lifting a projection.

Definition 1.3.6: The Exponential Map

Let $0 \rightarrow I \rightarrow A \xrightarrow{\pi} A/I \rightarrow 0$ be a short exact sequence of C^* -algebras and let $p \in M_n(\widetilde{A/I})$ be a projection representing a class $[p] - [s] \in K_0(A/I)$ (with s the scalar part of p). The *exponential map*

$$\partial_0: K_0(A/I) \longrightarrow K_1(I)$$

is defined by lifting p to a self-adjoint element $a \in M_n(\tilde{A})$ with $\tilde{\pi}(a) = p$ and then setting

$$\partial_0([p] - [s]) = [\exp(2\pi i a)] \in K_1(I)$$

The element $\exp(2\pi i a)$ is a unitary in $M_n(\tilde{A})$, and since $\tilde{\pi}(a) = p$ is a projection, $\tilde{\pi}(\exp(2\pi i a)) = \exp(2\pi i p) = 1$ (as $p^2 = p$ forces $\exp(2\pi i p) = 1 + (e^{2\pi i} - 1)p = 1$), so $\exp(2\pi i a)$ has entries congruent to 1 modulo I , hence lies in $\mathcal{U}_n(\tilde{I})$ and defines a class in $K_1(I)$.

The name records the formula: the connecting map in the even-to-odd direction literally exponentiates

a self-adjoint lift of a projection. The two maps ∂_0 and ∂_1 are the two connecting homomorphisms of the same six-term sequence, and under the dimension-shift isomorphism they are identified with each other applied to the suspended sequence, which is why a single periodicity statement governs both. The exactness proof below requires that both maps are well-defined homomorphisms, recorded first.

Proposition 1.3.7: The Connecting Maps Are Well-Defined Homomorphisms

The maps $\partial_1: K_1(A/I) \rightarrow K_0(I)$ and $\partial_0: K_0(A/I) \rightarrow K_1(I)$ of definitions 1.3.5 and 1.3.6 are well-defined group homomorphisms, independent of all choices of lift.

Proof:

Independence of the lift, index map. Suppose w, w' are two unitary lifts in $\mathcal{U}_{2n}(\tilde{A})$ of $\text{diag}(u^*, u)$. Then $w'w^* \in \mathcal{U}_{2n}(\tilde{A})$ projects to 1 under $\tilde{\pi}$, so $w'w^* \in \mathcal{U}_{2n}(\tilde{I})$. Writing $e = \text{diag}(1_n, 0)$, the projections wew^* and $w'e(w')^*$ are unitarily equivalent in $M_{2n}(\tilde{I})$ via $w'w^*$, hence equal in $K_0(I)$; and the difference with e is unchanged. Existence of a unitary lift: $\text{diag}(u^*, u)$ is connected to 1 by the rotation path of theorem 1.3.2, Step 1, applied in $\tilde{A}/I$; lifting that path of unitaries through the surjection $\tilde{\pi}$ (using that the unitary group surjects on identity components, a consequence of the fact that for b self-adjoint in $\tilde{A}/I$ any self-adjoint lift $\tilde{b}$ gives $\exp(i\tilde{b})$ lifting $\exp(ib)$) produces w with $\tilde{\pi}(w) = \text{diag}(u^*, u)$.

Independence of the representative, index map. If $u \sim u'$ are homotopic unitaries in $\mathcal{U}_n(\tilde{A}/I)$, lift the homotopy to a path of unitaries w_t in $\mathcal{U}_{2n}(\tilde{A})$; then $w_tew_t^*$ is a homotopy of projections in $M_{2n}(\tilde{I})$ (after subtracting e), giving equal $K_0(I)$ classes. Additivity: a lift of $u \oplus u'$ may be taken to be $w \oplus w'$ after reordering coordinates, and ∂_1 of a direct sum is the sum of the projection differences.

Exponential map. If a, a' are two self-adjoint lifts of the projection p , then $a - a' \in M_n(\tilde{I})_{sa}$, and the straight-line homotopy $a_s = (1 - s)a + sa'$ is self-adjoint with $\tilde{\pi}(a_s) = p$ throughout, so $\exp(2\pi i a_s)$ is a homotopy in $\mathcal{U}_n(\tilde{I})$, giving equal $K_1(I)$ classes. If $p \sim p'$ are homotopic projections, lift the homotopy to a path of self-adjoint elements and exponentiate to get a homotopy of unitaries. For additivity, $a \oplus a'$ is a self-adjoint lift of $p \oplus p'$, and $\exp(2\pi i(a \oplus a')) = \exp(2\pi i a) \oplus \exp(2\pi i a')$, whose K_1 class is the sum. Adding a scalar projection s changes a by a scalar self-adjoint with $\exp(2\pi i s) = 1$, leaving the class unchanged, so ∂_0 depends only on $[p] - [s]$. This establishes both maps are well-defined homomorphisms, as we sought to show.

With the connecting maps in hand, the half-exact sequences for K_0 and K_1 assemble into a single cyclic six-term sequence. This is the central computational tool of the subject: it lets one compute the K -theory of A from that of an ideal I and a quotient A/I , exactly as the long exact sequence of a pair computes cohomology in topology.

Theorem 1.3.8: The Six-Term Exact Sequence

Let $0 \rightarrow I \xrightarrow{\iota} A \xrightarrow{\pi} A/I \rightarrow 0$ be a short exact sequence of C^* -algebras. Then the diagram

$$\begin{array}{ccccc} K_0(I) & \xrightarrow{K_0(\iota)} & K_0(A) & \xrightarrow{K_0(\pi)} & K_0(A/I) \\ \partial_1 \uparrow & & & & \downarrow \partial_0 \\ K_1(A/I) & \xleftarrow{K_1(\pi)} & K_1(A) & \xleftarrow{K_1(\iota)} & K_1(I) \end{array}$$

is exact at every one of its six vertices, where ∂_0 is the exponential map (definition 1.3.6) and ∂_1 is the index map (definition 1.3.5).

Proof:

The proof has two layers. First, the horizontal three-term pieces (exactness at $K_0(A)$ and at $K_1(A)$) are the half-exactness of each functor, which we prove in full. Second, exactness at the four corners involving the connecting maps requires the index/exponential constructions; we prove exactness at $K_1(A/I)$ in full and at $K_0(A/I)$ in full apart from one path-lifting verification deferred to the references, then deduce the remaining two corners from the dimension-shift isomorphism of theorem 1.3.2 applied to the suspended sequence. The use of theorem 1.3.2 is legitimate and non-circular: that isomorphism was proved earlier and independently of this theorem (its injectivity and explicit inverse rest only on the framing construction of theorem 1.3.2, Steps 3a–3c, which never invoke the six-term sequence).

Step 1: Half-exactness of K_0 at $K_0(A)$. The composite $K_0(\pi) \circ K_0(\iota)$ is induced by $\pi \circ \iota = 0$, hence is 0, so $\text{im} K_0(\iota) \subseteq \ker K_0(\pi)$. Conversely, let $x \in \ker K_0(\pi)$. Represent $x = [p] - [q]$ with $p, q \in M_n(\tilde{A})$ projections having equal scalar parts. The hypothesis gives $[\tilde{\pi}(p)] = [\tilde{\pi}(q)]$ in $K_0(A/I)$, so after enlarging n and adding complementary projections there is a unitary $\bar{u} \in \mathcal{U}_n(\tilde{A}/I)$ with $\bar{u} \tilde{\pi}(p) \bar{u}^* = \tilde{\pi}(q)$. The doubled unitary $\text{diag}(\bar{u}, \bar{u}^*)$ lies in the identity component, hence lifts through $\tilde{\pi}$ to a unitary $w \in \mathcal{U}_{2n}(\tilde{A})$. Set $p' = \text{diag}(p, 0)$ and $q' = \text{diag}(q, 0)$, projections in $M_{2n}(\tilde{A})$ with $[p] - [q] = [p'] - [q']$. Then

$$\tilde{\pi}(wp'w^*) = \text{diag}(\bar{u}, \bar{u}^*) \tilde{\pi}(p') \text{diag}(\bar{u}^*, \bar{u}) = \text{diag}(\bar{u} \tilde{\pi}(p) \bar{u}^*, 0) = \tilde{\pi}(q')$$

so $wp'w^* - q' \in M_{2n}(I)$. Hence $wp'w^*$ and q' are projections agreeing modulo I with the same scalar part, and $[wp'w^*] - [q']$ is a projection difference over I , lying in $\text{im} K_0(\iota)$. Since $wp'w^*$ is unitarily equivalent to p' in $M_{2n}(\tilde{A})$, $[wp'w^*] = [p']$ in $K_0(A)$, so $[p] - [q] = [p'] - [q'] = [wp'w^*] - [q'] \in \text{im} K_0(\iota)$.

Step 2: Half-exactness of K_1 at $K_1(A)$. Identical in structure using unitaries: $K_1(\pi)K_1(\iota) = 0$ since $\pi\iota = 0$. For the reverse inclusion, let $[u] \in \ker K_1(\pi)$ with $u \in \mathcal{U}_n(\tilde{A})$. Then $\tilde{\pi}(u)$ is null-homotopic in $\mathcal{U}_n(\tilde{A}/I)$; lift the contracting path to a path of unitaries u_t in $\mathcal{U}_n(\tilde{A})$ (possible by the identity-component lifting used above) with $u_0 = u$ and $\tilde{\pi}(u_1) = 1$, so $u_1 \in \mathcal{U}_n(\tilde{I})$. Since $u \sim u_1$ in $\mathcal{U}_n(\tilde{A})$, we get $[u] = [u_1] \in \text{im} K_1(\iota)$.

Step 3: Exactness at $K_0(A/I)$. We show $\text{im} K_0(\pi) = \ker \partial_0$.

$\text{im} K_0(\pi) \subseteq \ker \partial_0$: if $x = K_0(\pi)(y)$ with $y = [r] - [s] \in K_0(A)$, $r \in M_n(\tilde{A})$ a projection, then r is itself a self-adjoint lift of $\tilde{\pi}(r)$, and $\exp(2\pi i r) = 1$ since r is a projection. Hence $\partial_0(x) = [\exp(2\pi i r)] = [1] = 0$.

$\ker \partial_0 \subseteq \operatorname{im} K_0(\pi)$: let $[p] - [s] \in K_0(A/I)$ with $\partial_0([p] - [s]) = 0$. Choose a self-adjoint lift $a \in M_n(\tilde{A})$ of p with spectrum in $[0, 1]$, possible because p is a projection: take any self-adjoint lift and apply the functional calculus with the function $t \mapsto \max(0, \min(1, t))$, which fixes the spectrum $\{0, 1\}$ of p and hence does not change $\tilde{\pi}(a) = p$. The strategy is to replace a by a self-adjoint lift a'' of (a stabilization of) p whose exponential $\exp(2\pi i a'')$ is exactly 1; such an a'' has spectrum in $\mathbb{Z}$, and the functional calculus with the indicator of $(1/2, \infty)$ then converts a'' into a genuine projection $r \in M_m(\tilde{A})$ with $\tilde{\pi}(r) = p \oplus 0$ (the scalar 0 part coming from the stabilization), whence $K_0(\pi)([r] - [s]) = [p] - [s]$ exhibits the class in the image. The hypothesis $\partial_0([p] - [s]) = 0$ says exactly that $\exp(2\pi i a)$ is null-homotopic in $\mathcal{U}_m(\tilde{I})$ after stabilizing, and it is this hypothesis that allows a to be corrected to such an a'' . The correction must be made multiplicatively, by deforming the unitary $\exp(2\pi i a)$ to 1 through a path of unitaries in $\mathcal{U}_m(\tilde{I})$ and lifting that path to a self-adjoint deformation of the lift; one cannot subtract a self-adjoint $b \in M_m(\tilde{I})_{sa}$ with $\exp(2\pi i b) = \exp(2\pi i a)$ and conclude $\exp(2\pi i(a \oplus 0 - b)) = 1$, since a and b do not commute and the exponential is not additive on non-commuting self-adjoints. The path-lifting argument is a several-page verification with no extra conceptual content beyond the constructions already given; it is deferred to [6, Chapter 12] and [1, Chapter 9].

Step 4: Exactness at $K_1(A/I)$. We show $\operatorname{im} K_1(\pi) = \ker \partial_1$.

$\operatorname{im} K_1(\pi) \subseteq \ker \partial_1$: if $[u'] = K_1(\pi)([v])$ with $v \in \mathcal{U}_n(\tilde{A})$, then $\operatorname{diag}(v^*, v)$ is a unitary lift of $\operatorname{diag}(\tilde{\pi}(v)^*, \tilde{\pi}(v)) = \operatorname{diag}((u')^*, u')$, and $\operatorname{diag}(v^*, v) \operatorname{diag}(1_n, 0) \operatorname{diag}(v, v^*) - \operatorname{diag}(1_n, 0) = \operatorname{diag}(1_n, 0) - \operatorname{diag}(1_n, 0) = 0$ in $K_0(I)$ (using $v^*v = 1$). Hence $\partial_1[u'] = 0$.

$\ker \partial_1 \subseteq \operatorname{im} K_1(\pi)$: let $[u]$ with $\partial_1[u] = 0$. Let $w \in \mathcal{U}_{2n}(\tilde{A})$ lift $\operatorname{diag}(u^*, u)$ and set $e = \operatorname{diag}(1_n, 0)$. The hypothesis says $[wew^*] - [e] = 0$ in $K_0(I)$, so after stabilizing wew^* and e are homotopic projections in $M_{2n}(\tilde{I})$ via a path f_t with $f_0 = wew^*$, $f_1 = e$. Lift this projection homotopy to a unitary path: there is $g_t \in \mathcal{U}(\tilde{I})$ with $g_0 = 1$ and $g_t(wew^*)g_t^* = f_t$, so $g_1wew^*g_1^* = e$, meaning g_1w commutes with e up to the ideal and, after adjusting, $g_1w \in \mathcal{U}_{2n}(\tilde{A})$ has $\tilde{\pi}(g_1w) = \operatorname{diag}(u^*, u)$ while commuting with e . A unitary commuting with $e = \operatorname{diag}(1_n, 0)$ is block-diagonal $\operatorname{diag}(x, y)$, so $\tilde{\pi}(x) = u^*$, whence $\tilde{\pi}(x^*) = u$ exhibits $[u] = K_1(\pi)([x^*]) \in \operatorname{im} K_1(\pi)$.

Step 5: The two remaining corners. Exactness at $K_0(I)$ (that is, $\operatorname{im} \partial_1 = \ker K_0(\iota)$) and exactness at $K_1(I)$ (that is, $\operatorname{im} \partial_0 = \ker K_1(\iota)$) follow from the previous four corners by the dimension-shift isomorphism of theorem 1.3.2. This appeal is not circular: theorem 1.3.2 is established before this theorem and independently of it (its injectivity and explicit inverse come from the framing construction of theorem 1.3.2, Steps 3a–3c, which never invoke the six-term sequence), so it is legitimate to use it here. Suspending the short exact sequence gives

$$0 \longrightarrow SI \longrightarrow SA \longrightarrow S(A/I) \longrightarrow 0$$

an exact sequence because the suspension functor $C_0(\mathbb{R}) \otimes (-)$ preserves exactness (tensoring a short exact sequence by the nuclear $C_0(\mathbb{R})$ is exact). The strategy is to read each ideal corner of the original sequence as a quotient corner of the suspended sequence under the natural identification $\theta_B: K_1(B) \cong K_0(SB)$. The three-term K_0 -exactness of the suspended extension at $K_0(SI), K_0(SA), K_0(S(A/I))$, together with θ , recovers the K_1 -row of the original (Steps 1–2 again), while the exponential and index maps of the suspended extension correspond under θ to the index and exponential maps of the original by naturality of θ (theorem 1.3.2, Step 4). Splicing the K_0 -six-term sequences of the extension and of its suspension along these identifications places the two ideal corners $K_0(I)$ and $K_1(I)$ of the original into positions where exactness is exactly one of the quotient-corner statements (theorem 1.3.8, Steps 3–4) for the suspended extension, which has already been proved. The diagram chase intertwining ∂_0, ∂_1 with θ across

the suspension is direct but lengthy bookkeeping with no content beyond the naturality already in hand; it is carried out in full in [6, Chapters 8–9] and [1, Chapter 9].

Exactness has now been established at all six vertices, completing the proof.

The six-term sequence collapses the entire homological machinery of the subject into a hexagon. Its power is leverage: knowing K_* of any two of I , A , A/I pins down K_* of the third up to extension data carried by the connecting maps. The two quotient corners worked out above, $K_0(A/I)$ and $K_1(A/I)$, show the mechanism transparently: an obstruction to lifting a projection (respectively a unitary) is exactly a class in $K_1(I)$ (respectively $K_0(I)$). The remaining two corners are the same statements read through the suspension, which is why no genuinely new argument is needed there.

Example 1.3.9: The Six-Term Sequence of a Quotient by a Hereditary Ideal

Consider the unitization sequence $0 \rightarrow A \rightarrow \tilde{A} \xrightarrow{q} \mathbb{C} \rightarrow 0$ for a non-unital A , where q is the augmentation $(a, \lambda) \mapsto \lambda$. This sequence splits via $\lambda \mapsto (0, \lambda)$, so all connecting maps vanish and the six-term sequence breaks into two split short exact sequences:

$$0 \rightarrow K_0(A) \rightarrow K_0(\tilde{A}) \rightarrow K_0(\mathbb{C}) \rightarrow 0, \quad 0 \rightarrow K_1(A) \rightarrow K_1(\tilde{A}) \rightarrow K_1(\mathbb{C}) \rightarrow 0$$

Since $K_0(\mathbb{C}) = \mathbb{Z}$ and $K_1(\mathbb{C}) = 0$, the split gives $K_0(\tilde{A}) \cong K_0(A) \oplus \mathbb{Z}$ and $K_1(\tilde{A}) \cong K_1(A)$. This recovers the definition $K_0(A) = \ker(K_0(\tilde{A}) \rightarrow K_0(\mathbb{C}))$ used for non-unital algebras and shows it is consistent with the six-term sequence: the splitting forces $\partial_0 = \partial_1 = 0$ exactly when the extension is split, the operator-algebraic shadow of the topological fact that a split cofibration contributes no boundary.

The dimension-shift isomorphism reduces all higher groups to K_0 of iterated suspensions, but on its own it does not close the cycle: a priori $K_0(S^2 A)$ could be a new group. Bott periodicity is the statement that the double suspension returns to the start. In topology this is $\Omega^2(BU \times \mathbb{Z}) \simeq BU \times \mathbb{Z}$ ([2]); the C^* -algebraic version is its exact translation under Gelfand duality (theorem 1.0.3).

Theorem 1.3.10: Bott Periodicity for C^* -Algebras

For every C^* -algebra A the *Bott map*

$$\beta_A: K_0(A) \longrightarrow K_0(S^2 A) = K_2(A), \quad \beta_A(x) = b \cdot x$$

given by external multiplication with the Bott generator $b \in K_0(S^2 \mathbb{C}) = \tilde{K}^0(S^2) \cong \mathbb{Z}$ (explicitly $\beta_A([p] - [s]) = [\beta(p)] - [\beta(s)]$ for the Bott projection $\beta(p)$ over S^2 defined in the proof) is an isomorphism, natural in A . Equivalently, via theorem 1.3.2, $K_1(SA) \cong K_0(A)$, and hence

$$K_n(A) \cong K_{n+2}(A) \quad \text{for all } n \geq 0$$

so the theory is 2-periodic with the only groups $K_0(A)$ and $K_1(A)$.

Proof Key ideas:

The proof is the analytic core of the subject and is genuinely beyond the elementary scope of this section; the full argument is given in the named references below, and the idea is sketched here per the deferral policy.

The Bott map. For a projection $p \in M_n(\tilde{A})$ one builds a projection-valued function on S^2

(equivalently on the double suspension) by the formula

$$\beta(p)(z) = \begin{pmatrix} p & 0 \\ 0 & 0 \end{pmatrix} + (\text{rank-one rotation in } z), \quad z \in S^2 = \mathbb{C} \cup \{\infty\}$$

concretely the Bott projection $\beta(p)(z) = \frac{1}{1+|z|^2} \begin{pmatrix} 1 & \bar{z} \\ z & |z|^2 \end{pmatrix} \otimes p + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes (1-p)$ over the Riemann sphere; this is the generator of $\tilde{K}^0(S^2)$ tensored with p . The map $[p] \mapsto [\beta(p)]$ is the external product with the Bott generator $b \in K_0(S^2\mathbb{C}) = \tilde{K}^0(S^2) = \mathbb{Z}$.

Why it is an isomorphism. For commutative $A = C(X)$ the statement is the topological Bott periodicity $\tilde{K}^0(X) \cong \tilde{K}^0(S^2X)$, which over a point is $\tilde{K}^0(S^2) \cong \mathbb{Z}$ and in general is [2] read through $\Omega^2(BU \times \mathbb{Z}) \simeq BU \times \mathbb{Z}$. For general A one proves the Bott map is an isomorphism by either (1) Cuntz's proof via the Toeplitz extension and the universal property of K_0 , reducing periodicity to the computation of K_* of the Toeplitz algebra (example 1.3.11), or (2) Atiyah's rotation-trick proof adapted to the operator setting, or (3) verifying the external product $K_0(A) \otimes K_0(S^2\mathbb{C}) \rightarrow K_0(S^2A)$ is an isomorphism by the structure of $K_0(S^2\mathbb{C}) = \mathbb{Z}$ and a Mayer-Vietoris / excision argument. The naturality and the multiplicativity of the external product reduce the general case to the scalar computation $K_0(S^2\mathbb{C}) = \mathbb{Z}$ established in example 1.3.4. The complete proof along route (1) is given in [6, Chapter 11], and along the operator-theoretic route in [1, Chapter 9].

Granting the scalar case and the multiplicativity of the external product, the general isomorphism follows, completing the argument.

Bott periodicity is what makes operator K -theory a genuine generalized cohomology theory in the sense of [5]: the two functors K_0 and K_1 , the suspension isomorphism, and the six-term sequence together satisfy exactness, homotopy invariance, and 2-periodicity, the operator analogues of the Eilenberg-Steenrod axioms ([2] records the topological side). The periodicity collapses the infinite ladder of connecting maps from successive suspensions into a single recurring hexagon, so that the index and exponential maps are, up to the Bott isomorphism, the only two connecting maps the theory ever produces. For the commutative algebra $C(X)$ the whole structure reduces, via Gelfand-Naimark duality, to topological K -theory: $K_0(C(X)) \cong K^0(X)$ and $K_1(C(X)) \cong K^1(X) = \tilde{K}^0(SX)$.

The Toeplitz extension is the worked example that makes every abstract construction of this section concrete: the index map becomes the Fredholm index, and the Fredholm index becomes the winding number. Let $H^2 \subseteq L^2(S^1)$ be the Hardy space, the closed span of $\{z^n : n \geq 0\}$, with orthogonal projection $P: L^2(S^1) \rightarrow H^2$. For $f \in C(S^1)$ the *Toeplitz operator* is $T_f = PM_f|_{H^2}$, where M_f is multiplication by f . The C^* -algebra $\mathcal{T}$ generated by all T_f is the *Toeplitz algebra*.

Example 1.3.11: The Toeplitz Extension $0 \rightarrow \mathcal{K} \rightarrow \mathcal{T} \rightarrow C(S^1) \rightarrow 0$

The commutator $T_f T_g - T_{fg}$ is compact for all $f, g \in C(S^1)$, and the compact operators $\mathcal{K} = \mathcal{K}(H^2)$ form a closed ideal of $\mathcal{T}$ with $\mathcal{T}/\mathcal{K} \cong C(S^1)$ via the *symbol map* $\sigma(T_f) = f$. This yields the short exact sequence

$$0 \longrightarrow \mathcal{K} \xrightarrow{\iota} \mathcal{T} \xrightarrow{\sigma} C(S^1) \longrightarrow 0$$

We compute its six-term sequence and identify the index map.

K-theory of the three terms. The compacts satisfy $K_0(\mathcal{K}) = \mathbb{Z}$ (generated by any rank-one projection, since $\mathcal{K} = \varinjlim M_n(\mathbb{C})$ and K_0 commutes with the colimit) and $K_1(\mathcal{K}) = 0$ (every unitary in $\tilde{\mathcal{K}} = \mathcal{K} + \mathbb{C}1$ is a compact perturbation of a scalar, connected to it). For the circle, $C(S^1)$

gives $K_0(C(S^1)) = \mathbb{Z}$ (generated by $[1]$, as every complex vector bundle over S^1 is trivial, $[2]$) and $K_1(C(S^1)) = \mathbb{Z}$ (generated by the class of the unitary $z = \text{id}_{S^1}$, since $\mathcal{U}_n(C(S^1))/\mathcal{U}_n(C(S^1))_0 \cong \pi_1(\text{U}(n)) = \mathbb{Z}$ detects the winding number of $\det$).

The six-term sequence. Substituting,

$$\begin{array}{ccccc} \mathbb{Z} = K_0(\mathcal{K}) & \xrightarrow{K_0(\iota)} & K_0(\mathcal{T}) & \xrightarrow{K_0(\sigma)} & K_0(C(S^1)) = \mathbb{Z} \\ \partial_1 \uparrow & & & & \downarrow \partial_0 \\ K_1(C(S^1)) = \mathbb{Z} & \xleftarrow{K_1(\sigma)} & K_1(\mathcal{T}) & \xleftarrow{\quad} & K_1(\mathcal{K}) = 0 \end{array}$$

The bottom-right entry is 0, so $\partial_0: \mathbb{Z} \rightarrow 0$ is zero and $K_0(\sigma)$ is surjective.

The index map is an isomorphism. The generator of $K_1(C(S^1)) = \mathbb{Z}$ is $[z]$. A lift of z to $\mathcal{T}$ is the unilateral shift $T_z = S$, which sends $z^n \mapsto z^{n+1}$ on H^2 ; it is an isometry, hence a partial isometry with $S^*S = 1$ and $1 - SS^* = P_0$, the rank-one projection onto the constants. By the partial-isometry formula of definition 1.3.5, for $v = S$,

$$\partial_1[z] = [1 - SS^*] - [1 - S^*S] = [P_0] - [0]$$

the class of a rank-one projection, which generates $K_0(\mathcal{K}) = \mathbb{Z}$; numerically $\partial_1[z] = +1$. The analytic Fredholm index of S is $\dim \ker S - \dim \ker S^* = 0 - 1 = -1$, so the index map records $[1 - SS^*] - [1 - S^*S] = [\ker S^*] - [\ker S]$, the negative of the Fredholm index, and $\partial_1[z] = -\text{index}(S) = w(z) = 1$. Hence ∂_1 is an isomorphism $\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$.

Consequences. Since ∂_1 is injective, exactness at $K_1(C(S^1))$ forces $K_1(\sigma) = 0$, and since ∂_1 is surjective, exactness at $K_0(\mathcal{K})$ forces $K_0(\iota) = 0$. Exactness at $K_0(\mathcal{T})$ then gives $K_0(\mathcal{T}) \cong K_0(C(S^1)) = \mathbb{Z}$ (generated by $[1_{\mathcal{T}}]$), and exactness at $K_1(\mathcal{T})$ gives $K_1(\mathcal{T}) = 0$ (it injects via $K_1(\sigma) = 0$ into $K_1(C(S^1))$ with image $\ker \partial_1 = 0$, and receives 0 from $K_1(\mathcal{K}) = 0$). Thus $K_0(\mathcal{T}) = \mathbb{Z}$ and $K_1(\mathcal{T}) = 0$: the Toeplitz algebra is K -theoretically a point, the operator-algebraic statement that $\mathcal{T}$ is the “cone” on the circle.

The Toeplitz index theorem. For an invertible $f \in C(S^1)$ (equivalently a class $[f] \in K_1(C(S^1)) = \mathbb{Z}$ given by the winding number $w(f)$ of f around 0), the Toeplitz operator T_f is Fredholm and

$$\text{index}(T_f) = \dim \ker T_f - \dim \text{coker } T_f = -w(f)$$

so that the index map records $\partial_1[f] = [1 - T_f T_f^*] - [1 - T_f^* T_f] = [\text{coker } T_f] - [\ker T_f] = -\text{index}(T_f) = w(f)$. The index map of the Toeplitz extension is therefore the winding number itself, which is the Gohberg-Krein / Toeplitz index theorem realized as the connecting homomorphism of the six-term sequence.

The Toeplitz extension is the keystone because it ties together three apparently separate notions: the abstract connecting map of homological K -theory, the Fredholm index of analysis, and the winding number of topology. Cuntz’s proof of Bott periodicity (theorem 1.3.10) runs through exactly this computation: $K_*(\mathcal{T})$ being that of a point is the engine that drives periodicity for all A via the tensored extension $0 \rightarrow \mathcal{K} \otimes A \rightarrow \mathcal{T} \otimes A \rightarrow C(S^1) \otimes A \rightarrow 0$. The same calculation, read over $C(X)$ instead of $\mathbb{C}$, is the Atiyah-Singer index theorem for Toeplitz operators with matrix symbol, where the winding number is replaced by the degree of the symbol’s homotopy class in $K^1(X)$.

Exercise 1.3.1

1. Let A be a unital C^* -algebra. Using the cone exact sequence $0 \rightarrow SA \rightarrow CA \xrightarrow{\text{ev}_1} A \rightarrow 0$ and the contractibility of the cone CA , deduce from the six-term exact sequence that $K_0(SA) \cong K_1(A)$ and $K_1(SA) \cong K_0(A)$. Explain how this recovers the dimension-shift isomorphism of theorem 1.3.2 and, combined with Bott periodicity, the relation $K_2(A) \cong K_0(A)$.
2. Compute K_0 and K_1 of $C(S^1)$ directly from the dimension-shift isomorphism and the values $K_0(\mathbb{C}) = \mathbb{Z}$, $K_1(\mathbb{C}) = 0$, using $C(S^1) \cong \widetilde{C_0(\mathbb{R})}$ and the split sequence $0 \rightarrow C_0(\mathbb{R}) \rightarrow C(S^1) \rightarrow \mathbb{C} \rightarrow 0$. Confirm your answer matches the values used in example 1.3.11.
3. Let S be the unilateral shift on H^2 . Verify directly that $S^*S = 1$ and $1 - SS^*$ is the rank-one projection onto the constant functions, and conclude that S is a Fredholm operator of index -1 . Then explain, using definition 1.3.5, why the class $[P_0]$ of this rank-one projection generates $K_0(\mathcal{K}) = \mathbb{Z}$ and equals $\partial_1[z]$.
4. Show that the exponential map $\partial_0: K_0(A/I) \rightarrow K_1(I)$ vanishes whenever the extension $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ splits by a $*$ -homomorphism $s: A/I \rightarrow A$ with $\pi \circ s = \text{id}$. Deduce that for the unitization sequence the six-term sequence breaks into two short exact sequences, recovering example 1.3.9.
5. Use the Toeplitz extension tensored with A , namely $0 \rightarrow \mathcal{K} \otimes A \rightarrow \mathcal{T} \otimes A \rightarrow C(S^1) \otimes A \rightarrow 0$, together with $K_*(\mathcal{T} \otimes A) \cong K_*(A)$ and $K_*(\mathcal{K} \otimes A) \cong K_*(A)$ (stability), to derive the periodicity statement $K_1(SA) \cong K_0(A)$. Identify which map in the six-term sequence becomes the Bott isomorphism.
6. Let $f \in C(S^1)$ be invertible with winding number $w(f) = 2$. Determine $\dim \ker T_f$ and $\dim \text{coker } T_f$ given that one of them is zero, and state the class $\partial_1[f] \in K_0(\mathcal{K}) = \mathbb{Z}$. Then describe how the answer changes for $w(f) = -3$.